SCIENCE / PHILOSOPHY

THE ARCHITECTURE OF REALITY

From space and time to causality, consciousness and existence

What the world is, what we can understand and what changes the way we describe it

Contents

A journey from the structure of the world to the limits and responsibilities of understanding.

Prologue. The Universe That Paused

Suppose the universe stopped for a billion years. Not just the stars, but the light between them. Not just hearts, but chemical reactions, clocks, memories in the process of forming. Then everything continued from exactly the state in which it had been interrupted. No trace of the pause remained inside.

Did anything happen?

Our first temptation is to answer yes: a billion years passed. But who counted those years? If time is what physical processes indicate, we have eliminated the very criterion by which the pause could have a duration. If time flows independently of any process, we have already assumed a particular metaphysics. And if someone outside the universe pressed a button, we have introduced another universe, with its own time, without explaining time itself.

This thought experiment does not demonstrate that time is an illusion. It does something more useful: it forces us to discover what we have hidden in the question.

The same thing can happen with space. Where is space, if every "where" presupposes it? With a cause: what produces the relationship through which one event produces another? With existence: in what way should existence itself exist? With understanding: how would we recognize a complete explanation of understanding without already using our capacity to understand?

Some difficulties arise because the world is complicated. Others arise because we try to use a tool outside the domain for which it was built. A ruler does not measure regret. But the conclusion is not that regret does not exist. Perhaps we need another measure; perhaps measurement captures only part of what we are trying to know.

This book follows a path. We begin with what seems most external: distances, durations, movements. We reach causality, meaning what connects changes. We then ask what the laws that make these connections stable are. We pass through life, emotion, and consciousness, where the world not only changes but seems to be experienced from within. Finally, we return to the question that encompasses them all: what exactly are we saying when we assert that something exists?

This path is not a ladder from simple things to mysterious ones. Space can be as enigmatic as pain. The difference is that, with space, we have grown accustomed to confusing competent use with ultimate understanding. We know how to find an address and assume we know what a place is. We know how to keep an appointment and assume we know what time is. We know how to say “I” and assume we know what the speaker is.

We will maintain an essential distinction: a concept can be constructed by the mind without its object being invented by the mind. A map is made by people; the mountain need not be. Sometimes, however, the object also depends on people: a financial debt is not a stone, but it can cause someone to be evicted from a house. At other times, we have an experienced reality, such as fear, which does not become fictional merely because it does not occupy space the way a table does.

Consequently, the question “real or imaginary?” will be broken down into more precise questions. Independent of whom? Real in what sense? Through what effects? With what criterion of identification? Would it exist without observers or merely without a particular observer? Can the error in a representation itself be a real event?

Neither “science has demonstrated” nor “science cannot explain” will serve as magic formulas. An observation can be robust while its interpretation remains disputed. A theory can explain much without explaining everything. A metaphysical hypothesis can be coherent without experimental support. And a question can be authentic even before we have an appropriate tool to investigate it.

There are also stakes beyond curiosity. Definitions allocate costs. Whoever defines efficient time can decide what another person’s waiting is worth. Whoever defines the cause of a failure can distribute blame. Whoever defines consciousness can expand or restrict the circle of beings whose suffering matters. Whoever defines understanding can build a school that produces autonomous people or people who repeat acceptable answers.

Asking who benefits does not mean declaring every idea a conspiracy. A true theory can be used unjustly; a false theory can console; an arbitrary convention can beneficially coordinate millions of people. Social interest alone does not establish truth. But truth does not cancel the obligation to examine its use.

This book's position is realism without triumphalism: we have serious grounds for assuming that we encounter a world that resists us, yet we have no right to identify that world automatically with the vocabulary through which we manage it. Understanding advances when an explanation risks losing to experience and when a definition can be revised without humiliating the people who used it.

Perhaps there is no final room in which all mysteries become obvious. But between possessing the final answer and being unable to say anything lies almost the whole of human knowledge.

On Evidence and Thought Experiments

The numbered references lead to the bibliography at the end. Historical works are used for their authors' ideas, not as validation of contemporary physics. Experimental results are separated from interpretations and speculative proposals. Invented numerical examples and small computational worlds are explicitly marked; they test reasoning within models rather than describe observations of the universe. The selected literature was checked for this edition in September 2026; the selection is indicative, not an exhaustive systematic review.

1. Where Is Space?

The Container, the Relation, and Geometry

Imagine that every object in the universe is moved simultaneously one meter to the right. No distance between them changes. No reference point is left behind. Has a displacement occurred, or have we merely changed the story we tell about positions?

The question separates two intuitions. The first sees space as a container: the objects can be removed, and the container remains. The second sees it as an order of relations: without things and without possibilities of relating, what would remain is no longer clear. Neither can be dismissed simply because the other seems more natural.

In Aristotle's physics, place was not the homogeneous, infinite space of modern textbooks. It was connected to the body surrounding another body and to the order of a world in which certain movements had natural directions. Asking where something was also meant asking where it belonged in the structure of the cosmos. Discussion of the void was thus also discussion of the possibility of movement and the nature of bodies. [1]

The transition to uniform mathematical space changes the type of question. We no longer seek the appropriate place for a thing, but measurable relations that can be the same anywhere. In the Newtonian reconstruction, absolute space provided a framework for true motion, distinct from motion relative to a chosen reference point. Leibniz, by contrast, maintained that space expresses the order of coexistences. A uniform displacement of the entire universe, without relational changes, raised for him the problem of a difference without sufficient reason. The controversy was not merely verbal: it involved how we identify acceleration and which entities we accept in an explanation. [2] [3]

It is worth observing the structure of the disagreement. Both thinkers could describe distances. The distance between two houses was not in dispute, but rather what makes a distance's existence possible. This difference between practical success and ontological interpretation will recur in every chapter.

Kant shifted the problem once again. Instead of deciding directly whether space is a container or a relation between things considered independently of us, he asked what conditions make our experience possible. In his project, space and time are forms of sensibility: ways in which things can be given to us. This does not mean that each person arbitrarily invents a universe. The distinction is between the world as it can appear to experience and things considered independently of the conditions of that appearance. [4]

We can use an analogy, with an important limitation. A camera produces images according to its construction, but it does not thereby produce the mountain photographed. The image's dependence on the device implies neither that the mountain is fictional nor that the image reproduces all its properties. The analogy does not prove Kant's philosophy. It merely shows why a representation's dependence on a cognitive system does not by itself resolve the problem of the reality represented.

General relativity introduced a more precise physical change: the geometry of spacetime is no longer merely the fixed stage of events but participates in the dynamic description of gravity. Distances and durations are linked to the metric structure, which is related to matter and energy through the theory's equations. It does not follow that all motion becomes indistinguishably relative or that a choice of coordinates can eliminate all gravitational effects. A change of labels does not erase, for instance, the physical differences between trajectories. [5] [2]

"Space is curved" need not be imagined as a sheet bent within a larger room. Curvature can be defined through internal relations: what happens to trajectories, to the transport of directions, to local measurements compared with one another. The image of the sheet helps initially and later becomes an obstacle if it forces us to invent an external space in which the entire universe would have to be placed.

The lesson is methodological. Understanding a concept may require abandoning the image that helped us learn it. A good metaphor is temporary support, not an obligation imposed on reality.

Can Space Emerge from Something Nonspatial?

Let us construct an imaginary universe consisting only of points and connections. We do not draw the points on a sheet; we retain only information about who can influence whom. Two points are “near” if a signal can pass between them in a few steps. They are “far apart” if many crossings are needed. In such a model, distance emerges from connectivity.

Have we already manufactured our space? No. We have constructed an analogue of one spatial property. The graph may have strange shortcuts, neighborhoods unlike three-dimensional ones, or distances that depend on the direction of travel. To obtain something resembling physical space, we must explain much more: dimensionality, locality, symmetries, signal behavior, and how geometry emerges at large scales.

This is precisely where quantum gravity programs in which geometry is not the primitive ingredient become interesting. In an influential theoretical argument, Mark Van Raamsdonk explored the connection between quantum entanglement and spacetime connectivity in holographic frameworks. The idea is not that two people thinking about one another create a geometric bridge. It is a mathematical relationship between quantum structures and gravitational descriptions, valid in specific contexts. [6]

Another family of proposals starts from sets of events with a discrete causal order. In the causal set approach, the elementary structure is not an ordinary network of boxes arranged in a given space, but a partial order with additional requirements. The program attempts to recover an appropriate continuous geometry when we look at large scales. However, not every discrete order produces a spacetime resembling the one observed. This is part of the difficulty, not a detail resolved by choosing the word “emergence”. [7]

These ideas justify an important intellectual possibility: what we use as an intuitive foundation might be a result. But possibility is not confirmation. We are not entitled to move from “there are models in which geometry is reconstructed” to “it has been demonstrated that our universe is made of information”. The latter statement is also too vague: what kind of information, about what, in what medium, and with what dynamics?

Let us return to the imaginary graph. For its inhabitants, who can send signals only through connections, operational distance would be real: it would limit communication and action. For the programmer, two internally distant nodes might be recorded in adjacent locations in the computer’s memory. The two descriptions do not cancel each other out. They concern different relations.

This offers a better answer than the opposition “fundamental or false”. A property can be real at the level at which it organizes possibilities for interaction, even if it depends on a deeper structure. The distance between two subway stations, measured by travel time, is not identical to the distance on the map. Neither is therefore a lie.

The difficult question becomes: which of these distances is relevant to the phenomenon being studied? A doctor, an urban planner, and a physicist can answer differently without any of them denying the others’ reality. The error arises when the answer suited to one purpose is proclaimed the only legitimate description of existence.

Space in the Brain and Space Shared Among People

The organism does not receive a neutral, ready-drawn map. Experiments by John O’Keefe and Jonathan Dostrovsky identified neural activity related to an animal’s position, while later work by Torkel Hafting and colleagues described cells with a regular pattern of spatial activation in the entorhinal cortex. These results made the study of spatial representation much more concrete. They do not demonstrate that external space is literally inside the brain or that it does not exist without it. [8] [9]

A navigation system needs a selective representation. For a moving being, traversable routes, obstacles, refuges, and the cost of returning may matter. A perfect description of every atom in the room would be useless if it did not allow the door to be found. This is a functional observation: a representation is also judged by the questions it enables the organism to answer.

It does not follow that evolution systematically favors falsehood. An error about the edge of a precipice can be costly. It follows only that a representation's usefulness does not require a complete metaphysics. You can have a good map for survival and a poor theory of the nature of space.

In society, the selection of relevance also becomes a distribution of power. Consider the same city as seen by a child, a wheelchair user, a property owner, and a person without a home. The buildings' coordinates are the same. Practical space is not. A step, a fence, a price, or a prohibition transforms geometric proximity into inaccessibility.

Distance in kilometers can omit the costs and risks of travel. The neutrality of the number does not guarantee the neutrality of the question.

A major social experiment offers an empirical reference point. Raj Chetty, Nathaniel Hendren, and Lawrence Katz's analysis of the Moving to Opportunity program found later benefits for children who had moved at younger ages to neighborhoods with less poverty, based on an intervention using vouchers distributed by lottery. For adolescents, the results differed and suggested possible costs of disruption. The study does not say that every move is beneficial. It shows that place can play a causal role in opportunities and that the timing of exposure matters. [10]

The question "who benefits from the concept of space?" therefore has several answers. Those who can navigate, build, coordinate, and verify measurements benefit. Those who control access and can present institutional distances as mere natural facts may benefit disproportionately. Those whose obstacles do not appear on the official map lose out.

Art can make such omitted spaces visible. Imagine two paintings of the same square. One preserves geometric perspective. The other enlarges the forbidden areas and compresses the places accessible to the protagonist. The second is less optically accurate and may be more existentially accurate. It does not compete with the cadastral plan.

2. What Passes When Time Passes?

The Clock Is Not Duration

A minute spent waiting for a medical result and a minute spent in a happy conversation may have the same measured duration and a very different experienced weight. Is time itself changing, or does only the way we inhabit it change?

The question seems to demand a choice between physics and experience. That choice is premature. We can distinguish the order of events, the interval measured by a clock, the direction of irreversible processes, and the sensation of passage. These are connected, but they are not synonyms. A physics book can explain differences between two clocks without explaining why a childhood afternoon seems more expansive than a recent week.

In Confessions, Augustine formulated the difficulty of time starting precisely from this familiarity that unravels when we try to explain it. The past is no longer present, the future is not yet here, and the present seems to shrink when we try to pin it down. He linked temporal experience to memory, attention, and expectation. This was not an experimental theory of the brain, but an analysis of the conditions under which we can experience and measure a succession. [11]

The power of the analysis becomes apparent in a melody. An isolated note is not the melody. To hear the musical phrase, we must somehow retain what has just passed and anticipate what might follow. The present of experience does not function like a mathematical point devoid of any extension. This does not demonstrate that physical time has a “psychological thickness”; it demonstrates that describing an experience requires relations between moments.

Bergson insisted on the distinction between lived duration and time treated as a succession of homogeneous units. Experiences can modify one another: a fear changes the meaning of a sound, and the ending of an event reorganizes how we understand its beginning. Counting segments does not exhaust this kind of continuity. Nevertheless, criticizing a spatialized representation of duration does not refute physical measurements. [12]

There are two symmetrical mistakes here. The first declares that only clock time is real and that everything else is an unimportant error. The second turns every subjective variation into evidence that individuals can physically alter time through willpower. Pain experienced as lasting longer does not automatically change the rate of an atomic clock. But the fact that the clock does not record suffering does not cancel suffering.

The experiment conducted by Chess Stetson, Matthew Fiesta, and David Eagleman on a controlled fall separated two hypotheses: a frightening event can be retrospectively estimated as longer without the person gaining better visual temporal resolution. The results supported this dissociation in the task studied. The sample and situation were limited; we do not have a universal law of all traumatic experiences. We do, however, have a useful warning: the sensation of slowing does not automatically prove faster perceptual processing. [13]

The same discipline of distinctions will also protect us when we discuss consciousness. What seems obvious from within can be authentic as experience yet misinterpreted as a mechanism.

Why Relativity Changes the Meaning of “Now”

In everyday life, we assume a shared present. Even if two clocks are not synchronized, we imagine there is a true time we could recover. Relativity restricts this intuition: the simultaneity of distant events depends on the reference frame, under the conditions specified by the theory. There is no universal procedure for transforming all clocks into a single privileged present for the entire universe. [5]

We can understand what is at stake without confusing the effect with the ordinary delay of light. Of course we see things after their signals reach us. But the relativity of simultaneity remains even after we methodically correct for delays in each frame. The disagreement concerns assigning times to distant events, not simply the fact that one observer received a photograph late.

The theory does not, however, say that every order is arbitrary. For events between which an ordinary causal influence can exist, spacetime structure imposes constraints. Freedom to choose coordinates does not provide freedom to turn any effect into its own cause. A flexible vocabulary can describe a world with very firm restrictions.

Proper time is the duration measured by a clock along its trajectory. Two clocks that separate and meet again can accumulate different durations, depending on their trajectories and gravitational conditions. This is not merely a metaphor for different life experiences. It is a measurable physical difference. [5]

In atomic-clock research, Tobias Bothwell and colleagues resolved the gravitational redshift even within an atomic sample of millimeter extent. Such a fine result moves the discussion from spectacular images of spacecraft into laboratory metrology. It tests a prediction of relativity, not a theory of time's personal meaning. [14]

From here to the idea of a "block universe", however, lies an interpretive step. We can represent events in a four-dimensional structure and discuss whether the past, present, and future have the same ontological status. The mathematical representation should not be confused with a photograph taken from outside time. The dispute between presentism, eternalism, and other views concerns what exists, not merely a convenient way of drawing a diagram.

Suppose we choose the interpretation in which all events are parts of a complete structure. Does it follow that effort is useless? No. Within that structure, effort may be precisely one of the processes through which the result emerges. Fatalism says that the result comes regardless of what we do. Determinism says, much more modestly and precisely, that what happens depends on the relevant states according to rules. The difference between "with my action" and "regardless of my action" is enormous.

This is a conceptual demonstration, not a solution to free will. It does, however, free us from a false alternative: either we are outside nature or deliberation has no function. A process can be natural and still modify what follows.

The Arrow, Memory, and the Right to Use Someone Else's Time

Why do we remember the past and not the future? Why does milk mix with coffee, while spontaneous separation would surprise us? Answers invoking entropy must be formulated more carefully than the slogan “everything tends toward disorder”. Entropy is not a universal measure of ugliness or chaos. In statistical descriptions, what matters is how many microstates are compatible with a macroscopic description and what initial conditions we assume.

An instructional image helps. We can arrange a deck of cards in a very specific order. There is only one order chosen in this way, but very many arrangements we would call “shuffled”. If the dynamics sufficiently explore the space of possibilities, we are much more likely to encounter an arrangement from the broad class than the exact special arrangement. We have not deduced the universe's thermodynamics from a deck of cards. We have identified the role of an asymmetry between coarse descriptions and their numerous realizations.

Boundary conditions are decisive. Murray Gell-Mann and James Hartle examined the relationship between time symmetry, cosmological conditions, and the arrows of time. A dynamical law alone does not exhaust the story of temporal direction. We must also ask what kind of state the system is in and what conditions are imposed at the ends of its history. [15]

This prevents a circular explanation: we cannot simply explain the special past by saying that the past is where entropy's increase begins. We have renamed the problem. The origin of a cosmological state that permits such asymmetries remains a distinct physical and philosophical question.

Memory adds another relationship: a present trace must carry information about something no longer present in the same way. A photograph, a fossil, and a neural memory are very different mechanisms, but all presuppose preserved correlations. If every change immediately erased all traces, there would be succession without accessible history.

Information also has material conditions. Antoine Bérut and colleagues' experimental verification of Landauer's principle investigated the connection between the logical erasure of information and thermal dissipation in a controlled system. The principle concerns particular operations and physical conditions; it does not mean that every act of thought costs exactly a universal quantity of energy. It shows why information processing should not be imagined as an activity completely detached from thermodynamics. [16]

Social time introduces another order, constructed yet effective. In his study of work discipline, E. P. Thompson traced the transformation of the relationship between activity, time measurement, and industrial organization. At stake was not the invention of physical time, but the consolidation of practices through which working time became monitored, negotiated, and valued. [17]

A standard hour can be liberating. Without coordination, meetings, transport, and remote cooperation become more difficult. But the same standardization allows someone's time to be purchased in blocks without noticing those blocks' differing costs. An hour at night, an hour of caregiving, and an hour of imposed waiting do not have the same meaning merely because they have the same duration.

Imagine an administration that saves ten minutes for each official by transferring thirty minutes of waiting to each applicant. In the internal report, efficiency rises. For the community, the total time consumed may increase. Physics has not been violated; moral accounting has been narrowed. This invented example shows why whoever defines the relevant unit can also decide who becomes invisible.

The cultural representation of time likewise has no single form. In a study conducted in Pormpuraaw, Lera Boroditsky and Alice Gaby found arrangements of temporal sequences along an east–west direction, not simply from left to right relative to the body. The result concerns the participants and tasks studied; it does not authorize a comprehensive psychology of a culture. It shows that some cognitive conventions considered self-evident are not universal. [18]

Art exploits precisely the separation between succession and presentation. A novel can show the effect first and the cause only afterward. A piece of music can turn expectation into tension and repetition into meaning. From such experiences, we can learn to distinguish the time of an occurrence from the time in which we understand it. Meaning need not be available simultaneously with the event.

Who benefits from measurable time? Almost everyone participating in complex cooperation. Who may lose? Those whose lived duration, recovery, or caregiving is reduced to a contextless box on a form. The solution is not to abolish clocks, but to refuse to make them the sole judges of life.

Time thus confronts us with a fruitful limit. We can measure some temporal relations very precisely without knowing why temporality exists. We can intensely experience passage without identifying its mechanism. And a civilization can become increasingly precise at timekeeping and increasingly inattentive to what it does with the time of those who constitute it.

3. What Does “Because Of” Mean?

Four Questions Hidden in a Single Word

A bridge has collapsed. The engineer speaks of a crack. The administrator, of an insufficient budget. A witness says the last truck was too heavy. A historian recalls an institutional culture of postponing repairs. Which of them has identified the cause?

Several may be right, but not in the same sense. Some describe the immediate mechanism. Others describe the conditions that made the mechanism likely. Others identify the intervention that would have prevented the disaster. If we turn the problem into a competition for a single word, we risk losing precisely the information needed to build a safer bridge.

Aristotle distinguished explanations through matter, form, the source of change, and purpose. In the example of a house, the materials, their organization, the builder, and the purpose of shelter answer different questions. We need not introduce these four explanations unchanged into current physics. It is worth retaining, however, the observation that “why?” does not always make the same request. [19]

When we ask why the heart beats, we may seek the mechanism of contraction, the organ’s evolutionary history, or the role of circulation in the organism. Explanation through function does not automatically imply that the future pulls the present along behind it. We can say that a system has a role without attributing human intentions to every process. Confusing function and intention is one source of the impression that nature must think before it produces order.

Hume challenged the idea of a causal necessity that we see directly. We observe sequences and regularities, then expect them to continue. But the leap from what has happened to what must happen is not justified merely by repeating experience. The problem was not that people should stop learning, but that the grounding of inference is more difficult than it seems. [20]

A thousand mornings on which a person feeds an animal can form a solid regularity from the animal's perspective. They do not guarantee the person's intentions the next morning. The example does not refute induction; it shows why a regularity becomes more informative when we understand the conditions sustaining it.

Kant responded by changing the level of analysis: causality is not merely a habit added to impressions, but participates in organizing the objective experience of succession. His project attempts to explain how we can have a world of events, not just a private stream of impressions. Nevertheless, a philosophical condition of experience does not directly yield the form of a particular physical equation. [4]

We still live within this tension today. We need causes in order to act, but causes do not come to us as visible labels on events. We must infer them, test them, and specify the alternative relative to which we consider them relevant.

Why a Correlation Can Tell the Story Backward

Let us invent a complete example, without claiming it describes a real study. An organization wants to discover whether a course improves examination pass rates. It has one hundred experienced employees and one hundred beginners.

Of the experienced employees, ninety take the course: seventy-two pass. Of the ten who do not take it, nine pass. The rates are 80% and 90%. In this group, course participants perform ten percentage points worse.

Of the beginners, ten take the course and two pass. Of the ninety who do not take it, twenty-seven pass. The rates are 20% and 30%. In this group, too, participants perform ten percentage points worse.

Now we aggregate the data. One hundred people took the course, of whom seventy-four passed: 74%. One hundred went without the course, of whom thirty-six passed: 36%. A sales presentation could announce a spectacular advantage for the course, even though in each of the two categories the association runs in the opposite direction.

The arithmetic is not contradictory. The aggregated groups have different compositions. The course attracted mainly people who already had a high chance of passing. But category-level comparisons do not by themselves prove that the course does harm either. There may be additional differences between participants and nonparticipants. The example demonstrates the insufficiency of a particular inference; it is not an experiment identifying a real effect.

The central distinction in causal inference is between observing an association and deliberately changing a condition. Judea Pearl formulated tools through which assumptions about structure can be expressed explicitly and linked to observational or interventional data. A causal graph is not a photograph of hidden causes. It is a set of commitments about relations on which the identification of effects rests. [21]

In a well-executed randomized experiment, intervention allocation reduces certain systematic confoundings. Questions still remain: who entered the study, what was measured, what side effects were omitted, how long did follow-up last, and in what other contexts does the result hold? Demonstrating an effect in one setting does not mean obtaining an unconditional law for all settings.

Causality also has a counterfactual dimension. When we assert that a fence prevented an accident, we compare the observed world with an alternative in which the fence was absent, holding other relevant conditions fixed. We do not directly observe both histories of the same event. We build the comparison's justification from mechanisms, experiments, regularities, and assumptions.

This dependence on alternatives explains why people can name different causes without being completely irrational. In a fire, oxygen is indispensable. Yet an investigation often looks for the short circuit, because it is the unusual and modifiable condition. If we were studying combustion in a spacecraft with a controlled atmosphere, oxygen concentration might become the central variable.

A useful explanation makes the choice visible: "a cause of what outcome, compared with what alternative, and with a view to which intervention?" When these questions remain hidden, the debate about causes can function as a struggle over the distribution of costs.

How Far Can Causal Order Go?

Ordinary experience suggests localized events influencing one another in a clear order. Quantum physics does not allow us to extend all these intuitions without caution. Experiments on Bell inequalities, including the one conducted by Bas Hensen and colleagues in 2015, challenge descriptions that simultaneously satisfy certain conditions of locality and independence of measurement choices. The result is neither a demonstration that mind creates matter nor a protocol for the controlled transmission of messages faster than light. [22]

We can understand the form of the lesson even without the experimental details. A theory can exclude a package of assumptions without telling us that every remaining extravagant assumption is true. When a suspect is eliminated, the first passerby is not automatically convicted.

Ognyan Oreshkov, Fabio Costa, and Časlav Brukner investigated a theoretical framework in which quantum correlations can be analyzed without assuming a fixed global causal order at the outset. This broadens physics' vocabulary. It is not equivalent to the everyday possibility of sending a warning into our own childhood and producing any paradox we wish. "Without a fixed global causal order" has a technical meaning; it does not mean the absence of all constraints. [23]

If causal order were itself emergent, we would have to explain why it is so stable at the scale of our lives. A strange foundation does not relieve a theory of its obligation to recover ordinary things. The very robust ordinariness of experience then becomes one of the most important facts to explain.

There is also a conceptual limit: not every dependence is a temporal cause. A triangle has three sides by virtue of what we mean by triangle, not because a definition struck an object and produced another side. A debt may exist by virtue of a contract and institutions. This constitution is not identical to the impulse transmitted between two balls. We will need such distinctions when we ask why the universe itself exists.

In social life, however, the most dangerous confusion is between explanation and absolution. If we identify conditions that contributed to an assault, it does not follow that the assault becomes acceptable. If we recognize an individual's choice, it does not follow that institutions have no responsibility. Accountability involves norms concerning capacity, control, prevention, and reparation; it cannot be read directly from an equation of motion.

Let us return to the bridge. An exclusively microscopic explanation can enumerate interactions up to the material's breaking point. It does not automatically answer who should have closed the bridge to traffic. An exclusively moral explanation can find someone to blame without identifying the defect that will destroy the next bridge. A competent society needs both types of work, connected but not confused.

Who benefits from causality's ambiguity? Sometimes those who can present a manipulable condition as an inevitability. At other times, those who attribute all a system's effects to one person. But structural explanations can also be instrumentalized: if "the system" is defined so that no one can do anything, an apparently profound analysis can become a technique of demobilization.

The practical criterion proposed here is simple: a good causal explanation should improve both prediction and the choice of interventions, without concealing its assumptions and the distribution of consequences. Sometimes it must be supplemented with a historical or moral explanation. Supplementation is not a defect of science; it recognizes that our questions are not all of the same type.

Causality is not merely the chain linking us to the past. It is the vocabulary through which we try to discover where there is still room for a difference produced by action. But if such differences are stable, what stabilizes them? Here we enter the territory of laws.

4. Who Writes the Laws of Nature?

A Law Is Not a Commandment

A stone cannot be fined for violating gravity. The observation sounds like a joke, but it reveals the trap in the word “law”. Social laws can be broken and remain laws. A purported physical law refuted under its conditions of application must be revised, restricted, or abandoned.

When we speak of laws of nature, we sometimes import the image of a legislator without noticing. Nature supposedly has instructions, and things execute them. This is a possible interpretation, not something demonstrated by an equation’s successful predictions. Another interpretation sees a law as a compact description of regularities. A third links it to things’ real capacities or dispositions. These families of positions attempt to explain the same stability without locating it in the same ontological place. [24]

First, the inventory must be separated out. There are dynamical laws describing changes. There are symmetry principles indicating transformations under which a description preserves its structure. There are conservation laws, constraints, and statistical relations. There are constitutive relations for particular materials and regimes. There are useful but fragile empirical regularities. And an initial condition does not become a law simply because we need it to calculate what follows.

In a simple pendulum model, the rule of motion and the position from which it is released play different roles. The rule can be the same across many experiments; the initial condition distinguishes them. If we replace the pendulum with a device subject to significant friction, we have not demonstrated that the universe has changed its fundamental law. We have exceeded the previous model’s idealizations.

The same caution applies to certain “laws” in economics or psychology. A regularity conditioned by institutions, learning, and expectations should not be treated as a cosmic constant. Sometimes people change their behavior precisely because they have learned the prediction. If an institution uses an indicator as a target, actors may learn to optimize the indicator rather than the phenomenon it was supposed to represent. This scenario can be deduced from the difference between purpose and measure; it requires no new natural force.

There are also logical or mathematical rules. In a proof, the conclusion is constrained by premises and rules of inference. This is not the same sense of constraint as in a physical experiment. The fact that an equation has a solution does not guarantee that nature realizes that solution. Its interpretation and connection to observations must be specified.

Therefore, laws do not directly "justify" the concepts of space, emotion, or existence. They constrain models of the phenomena to which we apply these concepts. The definition establishes what we are trying to discuss. The experiment checks whether the distinction made captures something reproducible. The theory explains relations. Metaphysics asks what kind of reality we assume. Research becomes confused when one of these tasks claims to have automatically resolved the others.

Symmetry, Necessity, and the Possibility of Change

One of the deepest connections discovered in mathematical physics is the one Emmy Noether formulated between certain continuous symmetries of the action and conserved quantities. The result has precise mathematical assumptions; it does not say that every kind of beauty or regularity produces a conservation law. In the appropriate context, invariance under time translations is connected to the conservation of energy. [25]

This connection also explains why we must be careful with slogans. In general relativity, local conservation and the existence of a global energy defined in the familiar way are not the same assertion. Geometry and the available symmetries matter. It is incorrect to use the idea that "the universe's energy is always a constant sum" without qualification as a solution to every cosmological question. [25] [5]

Mathematical beauty can thus have heuristic value: a symmetry suggests a structure to investigate. But beauty is not a vote requiring the universe to accept our theory. We can admire an equation and still demand a prediction that exposes it to the risk of being wrong.

What would it mean for a law to change? At least three distinct things. A system's effective behavior might change because it enters another phase. An apparent constant might change because, in a broader theory, it is the value of a dynamical field. Or the fundamental mechanism itself might change, if such a description makes sense and can be specified. We cannot deduce the third situation merely by observing the first.

Freezing water provides an accessible image of the first possibility. Macroscopic properties change dramatically without the atoms receiving a new cosmic constitution at that moment. Similarly, an effective theory can have parameters that depend on the scale at which we use it. Scale dependence is not identical to the temporal aging of laws.

Nor is testing constants purely verbal. To distinguish physical variation from a change of units, we must formulate operational comparisons and, particularly, examine dimensionless ratios. If we simultaneously change the ruler and what we call a meter, a numerical change may not express a detectable change in the world. The relevant question is which comparison between processes would yield a different result.

We can imagine a world in which one rule is periodically replaced by another. But if the sequence of replacements follows a stable rule, we have described a metalaw. If the sequence follows no accessible regularity, predictive capacity diminishes. In neither case have we automatically explained why the complete order exists. We have shifted the question by one level.

This shift is not necessarily useless. Many good explanations have shown that a phenomenon considered elementary results from another. It becomes misleading, however, when presented as eliminating all assumptions. A profound explanation can reduce the number of assumptions without reaching zero.

Where Might Order Come From?

Let us examine the answers without turning them into caricatured camps.

On a Humean view, laws belong to the best system for describing facts: a compromise between simplicity and explanatory power. We no longer need to imagine an external mechanism through which a law compels events. The difficulty shifts to the criteria for the "best" system and to whether regularity is merely recorded or actually explained. [24]

A conception of laws as principles governing the world takes necessity more seriously as an ingredient of reality. It answers the intuition that a law should also explain what would have happened in unobserved situations. Its cost is clarifying what this governance means without hiding in the word "necessity" the very mystery it sought to resolve.

A dispositional conception locates order in entities' powers: things behave this way by virtue of their nature. But here too the question returns: what identifies that nature, how do we know it independently of behavior, and why does it have precisely these powers? An answer can be coherent without being ultimate.

A structural perspective gives priority to relations. Instead of objects fully determined before any connection, we have entities identified by their place in a network of relations. This fits certain intuitions of modern physics, but the fit is not a metaphysical demonstration. What distinguishes a physically realized structure from one that is merely mathematically possible remains to be explained.

Theism can propose an intentional foundation for order. In some traditions, God continuously sustains existence; in others, the autonomy of created causes is emphasized. Occasionalism formulates a more radical variant, in which ultimate causal efficacy is attributed to God rather than to creatures considered separately. Islamic and Christian traditions have not held a single position on this issue. [26]

Serious criticism does not consist in declaring every religious answer meaningless in advance. It consists in asking what additional things it explains, what it assumes, and what observations might count against an associated empirical claim. If every possible outcome is compatible with the same unspecified divine intention, the hypothesis may function as a metaphysical interpretation, but it does not discriminate between experimental outcomes.

Emergence offers another direction. Ted Jacobson showed how Einstein's equation can be obtained, in a theoretical argument, from thermodynamic assumptions about local horizons and entropy. This is an important example of a change in an equation's status: it can be viewed as an equation-of-state relation, not necessarily as an ultimate microscopic instruction. The argument has assumptions and a domain; it does not prove that we have identified the universe's final substrate. [27]

Selection among possible worlds or cosmological regions raises the observer problem. We should not be surprised that we observe conditions compatible with the existence of observers. But this observation alone does not produce a probability for different laws. That would require specifying the alternatives, their distribution, and how observers are counted. Without these ingredients, invoking selection may merely elegantly rephrase our surprise.

There remains the possibility of a brute fact: this order exists, and we have no deeper explanation. It is frustrating, but not incoherent simply because it is frustrating. Conversely, an answer's giving us satisfaction is not evidence that the universe chose it. The human need for closure is a fact about people, not a mandatory rule of reality.

Who benefits from the image of fixed laws? Research, coordination, and trust in reproducible interventions. But the rhetoric of inevitability can also protect changeable social arrangements. Who benefits from the image that everything is malleable? Creativity and reform, sometimes; at other times, the seller of unconstrained promises. Neither rigidity nor fluidity has moral value independently of its object.

A mature society should be able to say simultaneously: this physical constraint is real; this institutional rule is negotiable; this limit of the model is unknown. Mixing them produces either fatalism or magic. Separating them creates room for knowledge and action.

5. More Than the Sum of the Parts?

Three Meanings of Reductionism

If we knew every particle of a library perfectly, would we also know the meaning of its books? The question is less simple than it seems. A sufficient physical description might determine every letter. But reading the letters, identifying the language, and understanding irony require other organizations of information. Dependence on a medium is not identical to the accessibility of an explanation.

“Reductionism” can mean at least three things. Ontologically, it can assert that we need no additional type of substance to explain complex objects. Methodologically, it can recommend breaking systems into components to understand mechanisms. Explanatorily, it can claim that a lower-level explanation replaces all higher-level explanations. The first two do not automatically imply the third. [28]

Take an ordinary statement: the window broke because a stone hit it. We can descend toward elasticity, material structure, and microscopic interactions. This deeper analysis is valuable. Yet if we want to prevent another act of vandalism, a more precise description of atomic bonds does not replace investigating motives, surveillance, or conflict.

This does not bring forth two disconnected worlds. It brings forth different questions addressed to the same world. Reduction can explain a phenomenon’s possibility without by itself supplying the most useful explanation for all its aspects.

Philip Anderson condensed the critique of naive reconstruction into the formula “more is different”. Knowing the laws of components does not automatically provide an effective understanding of collective organization. In a later result, Mile Gu and colleagues demonstrated general limits on deriving certain macroscopic properties for an idealized class of infinite periodic Ising lattices. The qualification “infinite” is essential: the result does not show that every property of every finite system is impossible to calculate. [29]

We thus have a scale of difficulties. Sometimes calculation is possible and cheap. At other times, it is possible in principle, but resources make it impractical. Sometimes a problem formulated for an unbounded class of systems admits no general algorithm. And there are poorly formulated questions in which what is missing is not calculation but the criterion for the result. Calling all these situations "mystery" does not help.

At the opposite extreme, the word "emergence" can conceal the absence of an explanation. If we say a property arises from interactions, we must indicate which interactions, what organization, under what conditions, and what difference changing them makes. Otherwise, "emergent" becomes merely the respectable form of "and then something happened".

A more precise mechanism than a general appeal to emergence is renormalization. In studying critical phenomena, the description changes gradually from small details to larger scales. Some microscopic differences lose their importance for the behavior under investigation; others remain decisive. Systems with different components can thus belong to the same universality class. They have not become identical in every respect: they share certain relations and forms of behavior in the relevant regime. The idea of fixed points of this transformation between scales explains why collective regularities can be robust. [30]

For a controlled analogy, imagine a mosaic viewed first up close and then from afar. In the first case, we see every piece. In the second, we retain only averages over blocks. If we change one piece, the overall image may remain almost the same. If we change the organization across large regions, the difference persists. This averaging is not itself a derivation of the physical theory, but it shows why "losing information" can enable us to find a relevant regularity, rather than merely degrade knowledge.

There is a price, however. Two very different microscopic configurations can produce the same collective description. The success of the macroscopic law then does not uniquely identify the substrate. The paradox is instructive: precisely because an effective law ignores many details, it can be useful, but also insufficient to reconstruct all those details. Robust prediction does not imply a transparent foundation.

This is a partial answer to the question of how laws arise: some regularities are stable because numerous microscopic realizations converge toward the same description at the observed scale. The question remains why the family of mechanisms allowing this convergence exists. We have explained the autonomy of a collective law without claiming to explain the existence of every possible law.

How the Whole Can Matter Without a Magical Force

Imagine two devices built from the same components. One is wired as an alarm, the other as a musical instrument. Their inventory of parts may be identical; their behavior is not. The difference is not a new substance added to the parts. It is the organization of relations among them and with the environment.

In such a case, explaining only the parts means omitting precisely the variable that distinguishes the systems. Conversely, speaking only about "the whole" without showing how it is realized in components leaves the explanation suspended. A good analysis moves between levels.

Constraints provide an example. A channel's shape changes how water flows. We can call this an influence of the macro level on local movements without assuming that shape acts outside physics. The shape is physically realized by walls; its global description selects the relevant aspect. "Top-down" can designate an explanatory relationship between levels of organization, not an immaterial order transmitted to molecules.

In biology, a mutation has no effect independent of the organism and environment. An organ does not function in a vacuum. And organisms sometimes modify their environment, changing the conditions under which other processes will act. These functional observations prevent us from simplistically identifying the cause with the smallest part we can name.

Natural selection, in the basic form Darwin formulated, connects variation, inheritance, and differences in reproductive success. It presupposes no mind planning adaptations, nor does it provide a moral ranking of beings. What persists depends on conditions and history; it is not automatically what a designer concerned with everyone's welfare would choose. [31]

Biological success is not the same as economic success, and neither is identical to a good life. We can imagine behavior that brings immediate resources but later destroys essential relationships. We can also imagine individually costly behavior that sustains a cooperative environment. To say what is selected, we must describe the mechanism of transmission and the relevant horizon, not merely admire the visible winner.

Let us construct a small imaginary game. Two participants can cooperate or betray. If both cooperate, each receives three units. If one betrays and the other cooperates, the first receives five and the second zero. If both betray, each receives one. In a single round, the incentive structure favors betrayal. But if the interaction repeats, the two can condition their behavior on the past. The immediate two-unit benefit of betrayal may be outweighed by the loss of future cooperation.

We have not demonstrated that people are good or bad. We have shown that the same capacity can be advantageous or disadvantageous depending on how relationships are organized. Asking whether empathy is "biological self-sabotage" without specifying the environment, partners, and transmission of traits is therefore like asking whether speed is good without saying which road we are traveling.

This distinction has a political consequence. Institutions do not merely choose between already formed people; they can change which behaviors are rewarded, observed, and imitated. It does not automatically follow that cultural changes immediately produce genetic changes. Transmitting a rule, selecting a habit, and changes in genetic frequencies are distinct processes that must be investigated separately.

When Simplicity Produces Complexity and When It Does Not

A cellular automaton is a mathematical world in which very simple elements update their state according to their neighbors and a local rule. Matthew Cook demonstrated that rule 110 can support universal computation in his construction. This result shows that a short rule can permit very rich computational behaviors. It does not show that every random starting state produces intelligence or that our universe uses exactly this rule. [32]

The difference between "permits" and "usually produces" is decisive. A box of letters permits writing a treatise; shaking the box is not a sufficient method of writing one. A dynamics may contain the possibility of complex computation, but that computation's emergence and maintenance may require very special initial conditions, resources, and structures.

When someone proposes that life or consciousness arises from a few simple rules, the right question is not "how can something so small produce something so large?" The example of universal computation shows that simplicity of the rule does not exclude richness of behavior. Better questions are: which rules, which initial states, what selection of persistent structures, what energy sources, what criterion of life or consciousness?

There is another trap. We draw a moving pattern in a simulation and call it a "particle". This can be a useful abstraction if the pattern persists, interacts, and permits predictions. But we have not demonstrated that it possesses all the properties of an actual physical particle. The name must be earned through structure and behavior, not merely visual resemblance.

The same criterion applies to a simulated emotion. If an agent avoids danger, we have implemented avoidance behavior. If it adjusts its internal resources, we have implemented regulation. If it reports "I am afraid", we have also implemented a report. Whether all of this constitutes experienced fear is a further question, dependent on a theory of experience and evidence that simply naming it does not provide.

In society, reductionism can be liberating when it dismantles mystifying explanations and discovers modifiable mechanisms. It can become oppressive when it reduces a person to a score and treats what the score omits as nonexistent. Holism can correct this blindness, but it can also protect authorities invoking a “whole” inaccessible to any verification.

There is therefore no automatic moral choice between small and large. What matters is transparency in moving between levels. A hospital needs biological mechanisms, organization, and the patient’s experience. An explanation connecting these levels is stronger than one proclaiming the definitive victory of just one.

Art offers an analogue without standing in for physical evidence. The same vocabulary can form an insult, a consolation, or a poem. Order and context alter the effect. Meaning is not an additional ingredient poured over letters, but neither can it be recovered by simply counting them. Precisely what must be explained is the relationship between medium, organization, and interpretation.

The reasonable position is neither “everything reduces” nor “nothing important reduces”. It is more demanding: every explanation must show what it preserves when changing levels and what it loses. If it loses the property it was supposed to explain, it has not achieved a complete reduction. If it invokes additional properties, it must show why they are necessary and how they could be recognized.

With this requirement, we can move from the world’s order to its experienced interior. We no longer ask only how matter moves. We ask how a part of nature comes to have something to gain or lose and, perhaps, how that gain or loss comes to be felt.

6. Why Is There Something That Matters to Someone?

From Passion to the Regulation of a Life

A photograph lies on the table. For one observer, it is a surface of contrasts and colors. For another, it is the last image of a loved one. No analysis of pigments, taken alone, explains why the same object can be indifferent or devastating.

We need not choose between saying that the photograph physically possesses sadness and saying that sadness is unreal. The relationship matters. The object encounters an organism with memory, attachments, anticipations, and a particular history. The experience is not only in the photograph, and it cannot be fully understood without the photograph.

We will use several working distinctions, not definitions proclaimed universal. "Affect" will designate general dimensions of experience, such as pleasant–unpleasant and arousal–calm. "Emotion" will designate an episode organized in relation to something significant, involving variable combinations of appraisals, body, action tendencies, and experience. "Affective experience" will designate the felt aspect. "Sentiment", as used here, can designate a more enduring orientation, such as love or resentment. Everyday language does not always respect these boundaries, and researchers do not share a single vocabulary.

The distinction helps us avoid an error: love does not cease to exist at every moment when the loved person is not the focus of attention. A relational disposition can persist without a permanently intense episode. Conversely, an intense episode does not by itself establish a lasting commitment.

In Ethics, Spinoza treats affects as part of nature and links them to variations in our power to act. Instead of considering them merely shameful accidents of a mind that ought to be pure, he tries to make them intelligible through causes. Freedom is not the absence of all determinations, but is connected to a more adequate understanding of them. We can find a fruitful perspective here without adopting his entire metaphysical system. [33]

Darwin brought emotional expression into a natural history that includes continuities between humans and other animals. William James later challenged the intuition that a complete emotion arises first and only then does the body execute its command. In his analysis, perceiving bodily changes played a constitutive role in what we call emotion. Not all current research retains James's theory, but the question of the body's role can no longer be treated as a detail. [34] [35]

These shifts have a common consequence: emotion ceases to be merely reason's adversary. An organism needs to select what is urgent, what is worth preserving, and what can be ignored. An emotion's having a role does not mean its verdict is always correct. An alarm can be indispensable and still produce false alarms.

A perfectly indifferent system would have a problem different from error: why would it choose to do anything? Even a machine without experience can be given objectives and evaluation functions. This shows that prioritization can be defined functionally. It does not yet show why, in some organisms, prioritization is accompanied by experience.

Here the bridge between emotion and consciousness appears. Some questions concern behavioral control: how do we avoid danger? Others concern experience: why can avoidance hurt before contact with danger? If we confuse them, we will claim too quickly either that we have explained experience or that we have explained nothing.

Are Emotions Discovered or Constructed?

Take two people with racing pulses before going onstage. One calls the state panic, the other excitement. We cannot conclude that their states are identical simply because one bodily indicator is similar. But neither can we conclude that the verbal label creates their bodies, situation, and histories out of nothing.

Basic emotion theories seek relatively distinct biological organizations, with recurring functions and patterns. Appraisal theories emphasize how a situation is interpreted in relation to goals. Constructionist theories examine how episodes are assembled from more general processes, concepts, experience, and regulation of the organism. These approaches do not overlap perfectly, nor do they represent a simple choice between biology and culture. [36]

In the theory of constructed emotion, Lisa Feldman Barrett proposes a central role for interoception, prediction, and categorization. It is an influential theoretical model, not the field's final verdict. It invites us to ask which regularities arise from the organism's organization and which depend on the categories through which situations are recognized. [36]

A study by Alan Cowen and Dacher Keltner identified twenty-seven categories of emotional experience, connected by gradual transitions, in self-reports elicited by a particular set of videos. The number should not be turned into a definitive periodic table of the soul. It depends on the material, participants, reporting, and analysis. The result is interesting precisely because it suggests an organization richer than a few isolated and completely separate labels. [37]

We can thus formulate a general principle: discovering a statistical structure does not by itself establish the ontological status of the categories. If the data form three groups, we must ask why: three distinct mechanisms, three contexts, three frequently used words, or an effect of the measuring instrument? Grouping is the beginning of explanation, not its end.

The same applies to facial expressions. A smile can occur in joy, in an attempt at politeness, or in a situation of discomfort. The example logically shows why an expression considered without context does not provide an infallible identification of a state. A system classifying muscle movement has not thereby demonstrated that it reads the person's experience.

Constructing a concept can nevertheless change experience. Suppose someone has long felt a tension they treat as a personal flaw. They then discover a distinction between fatigue, fear, and humiliation. The new distinction may alter their questions and actions. But a label's benefit should be judged by what it enables us to observe and do, not by how much it appears to explain instantly.

There are also labels that narrow. If every dissatisfaction becomes "negativity", we can no longer distinguish a disproportionate reaction from a justified protest. If every fear becomes pathology, we may lose information about danger. If every suffering becomes evidence of superior depth, we may romanticize a state that needs help. A name can liberate or confine, depending on its use.

Emotions are therefore neither simple internal objects with perfect natural labels nor bodiless social fictions. At least for practical analysis, they must be followed as situated processes: organism, history, interpretation, environment, and relationship. A complete explanation should specify each one's contribution and show what changes when we intervene.

Who Is Allowed to Feel and Who Must Smile?

The sociology of emotions adds a question the laboratory may omit: which emotion is permitted to whom? Arlie Hochschild analyzed "feeling rules" and the work through which people try to shape their experiences according to social expectations. We do not manage only visible behaviors; sometimes we are asked to become, inwardly, the person the role demands. [38]

Imagine two scenes. A customer declares indignation and demands immediate attention. An employee, equally indignant, must respond calmly and warmly. Neither need have a more rational biology than the other. Their roles distribute differently the right to express emotion and the obligation to regulate it.

These rules can serve good functions. A teacher must not unload every irritation onto a student. A doctor must be able to continue helping in a difficult situation. But an ideal of self-regulation can become exploitation when an institution demands unlimited affective availability without recovery time, recognition, or protection.

Who benefits from the difficulty of defining emotion? A service provider may profit by calling any automatic classification “emotional understanding”. A leader may profit by calling the discomfort caused by their own decisions “a lack of resilience”. Conversely, marginalized groups may benefit when a new vocabulary makes previously denied experiences describable. These are possible mechanisms, not general accusations against everyone using these terms.

Economic success can sometimes reward the correct expression more than authentic experience. But authenticity is no simple solution either. A sincere outburst can hurt; disciplined kindness can protect. The ethical question is not only whether the emotion is “true”, but whether the relationship in which it is demanded preserves the autonomy and dignity of those involved.

At the biological level, emotion’s costs must be judged contextually. Fear can prevent dangerous exposure and hinder necessary exploration. Attachment can sustain caregiving and make separation painful. We need not proclaim every suffering adaptive to acknowledge that an affective system has functions. A mechanism can have a history of selection and become ill-suited to a changed environment.

Art offers a space in which the relationship between emotion and action can be reconfigured. We can experience sadness in a story without wishing the story would disappear. We can explore fear without being in the danger represented. This does not make the emotion false; it changes its conditions and meaning. Represented suffering may be tolerable precisely because our relationship to it differs from immediate life.

Compare two imaginary songs about loss. One directly names sadness. The other builds an interrupted rhythm and leaves an expectation unfulfilled. The first conveys a category; the second can organize an experience. Understanding artistic emotion does not mean merely translating every sound into a proposition. It also means observing what form does to attention and anticipation.

The remaining question is radical: what distinguishes a system that merely avoids a state from one for which that state is actually bad?

7. Who Is Here When We Say “I”?

Experience, Access, and the Imaginary Observer

Imagine a device that sees a flame, withdraws, explains the danger, and says it was frightened. Now imagine you can inspect each of its operations. You find sensors, memory, evaluations, decisions, and sentences. What else would you have to find to say that someone is there?

The question is not resolved by requiring an inner chamber in which a little observer watches the images. That observer would, in turn, need an explanation of seeing. If we repeat the construction, we do not explain experience; we move it from one room to another.

“Consciousness” brings together phenomena we must distinguish. Wakefulness is not identical to every conscious content. Access to information for reporting and control is not, by definition, identical to the experienced character of that information. Self-consciousness need not be assumed in every experience. And the ability to speak about consciousness is an additional competence, not a neutral definition of all its forms.

David Chalmers formulated the now-famous distinction between functional problems and the hard problem of experience: why are processes that discriminate, integrate, and control accompanied by something experienced? The expression “easy problems” does not mean those mechanisms are simple to discover. It means we can see the form of a functional explanation more clearly. The status of the explanation of experience remains disputed. [39]

Take the color red. We can ask which wavelengths are present, how receptors respond, what information becomes available for choice, and which word is used. Every answer adds knowledge. The question of how red appears to the subject should not be eliminated merely because it is harder. But neither should it be treated as evidence that mechanisms have no relevance.

The physicalist position asserts, in different forms, the mental world's dependence on physical reality. It should not be confused with the idea that neurons are classical balls or that psychological vocabulary is useless. Its challenge is to clarify in what sense a sufficient physical description would also explain experience, not just associated behavior. [28]

An opponent may answer: however many relations we describe, they do not tell us why experience exists. The physicalist may reply: this impression of distance might arise from the difference between our ways of accessing the same process. Neither the reply nor the objection automatically becomes an experiment. We need more precise concepts and predictions linking interpretations to concrete organizations.

A similar problem arises when we try to identify experience in someone unable to respond in the expected way. The absence of a report may mean the absence of experience, but it may also mean the absence of access to the reporting channel. This is a logical argument about measurement, not a sufficient clinical criterion. It would be wrong to decide a patient's condition from a thought experiment.

Conversely, the existence of a report does not settle everything by itself. A system can be trained or built to emit certain statements. What we need is convergence between reports, behaviors, mechanisms, and interventions, not the selection of a single sign turned into an oracle.

Theories Willing to Be Challenged

Contemporary theories do not all seek the same thing in the same place. The global workspace family emphasizes making information widely available to different processes. Recurrent processing theories give importance to interaction loops, not merely one-way transmission. Higher-order theories investigate the relationship between a state and its representation as a state of the subject. These ideas can inspire tests, but they must be formulated precisely enough for a result to count against a version of them. [40]

Integrated information theory, in Larissa Albantakis and colleagues' IIT 4.0 formulation, starts from properties attributed to experience and proposes an identity with a system's intrinsic, integrated, irreducible causal structure. It does not simplistically assert that any computer with abundant data becomes conscious. What matters is the causal organization described by the theory. The transition from phenomenological premises to the proposed physical identity is, however, a theoretical commitment, not a truth established merely by definition. [41]

An adversarial study published by the COGITATE consortium in 2025 compared predictions from IIT and global neuronal workspace theory. It involved 256 people, with measurements using different techniques. The results supported some predictions and challenged important aspects of both theories. The tests mainly targeted neurobiological implementations and operationalized predictions, not the direct proof or refutation of their entire mathematical or philosophical cores. [42]

The significance of the result is not that research failed. It is that two families of ideas agreed to specify in advance what should be observed. Where a theory can be retrospectively compatible with anything, experience has little selective power. Where criteria are established before the result, disagreement can become productive.

Nevertheless, adversarial collaboration is no magical solution either. Two theories can share a mistaken assumption. A test can miss the relevant difference. An indicator can measure access to reporting more than experience. The lesson is to preserve the architecture of criticism, not to treat an experimental format as a guarantee of truth.

We can understand the requirement through an imaginary example. We have two agent models. In the first, a message is transmitted to all subsystems; in the second, information remains local, but the system has memorized appropriate verbal responses. On repetitive questions, the agents may seem identical. When a new task requires coordination between memory, planning, and perception, they may diverge. We have tested a functional difference. We have not deduced, from this test alone, which one has experience.

This modesty matters for artificial intelligence. Patrick Butlin and colleagues' report proposes indicators derived from several theories of consciousness. The indicators are investigative tools, not a universal certificate. Assessment depends on the theories' validity and knowledge of the system's architecture. Conversational fluency, considered alone, cannot close the dispute. [40]

Nor do we have a shortcut in the opposite direction: "it is artificial" does not logically demonstrate "it cannot feel". Such a conclusion would require showing which necessary property is missing and why it cannot be realized. An object's origin does not automatically determine all its capacities.

What Do We Pay for Each Metaphysics of Mind?

Dualism treats the mental and physical as fundamentally distinct in a relevant sense. Its intuitive advantage is that it does not reduce experience to what seems not to contain it. The difficulty is explaining the relationship: how do they interact, what observable differences result, and why is mental life so closely linked to bodily organization? Naming two domains does not explain the bridge between them.

Idealism gives the mental a fundamental role in what we call reality. Not all its forms maintain that individual desires produce the world. Some try to explain the shared order through a mental foundation transcending the person. The cost is recovering the world's practical independence, intersubjective agreement, and the success of physical descriptions without turning all these into mere indistinguishable assertions. [43]

Panpsychism proposes, in the variants relevant here, that elementary mental aspects are fundamentally widespread in nature. It should not be caricatured as a theory of jealous electrons. It tries to avoid experience arising from something completely devoid of any experiential aspect. The combination problem remains severe: how would many such aspects constitute a single subject rather than a mere collection? Adding experience to the ingredients does not automatically explain the result's unity. [44]

Neutral monism and dual-aspect views attempt another move: perhaps “mental” and “physical” are partial ways of describing a foundation exhausted by neither. Spinoza provides a historical reference for the idea of a single reality conceived under distinct attributes. The advantage is avoiding two completely alien worlds; the difficulty is specifying the relationship without unity remaining merely a name. [33]

Illusionism, defended by Keith Frankish, challenges a particular conception of phenomenal properties and proposes that introspection misleadingly represents complex functional states. It should not automatically be translated as “no one suffers”. The dispute concerns the nature attributed to experience. Critics ask whether explaining the impression already presupposes the very experience to be explained; defenders try to formulate the impression without that commitment. [45]

These options are not experimentally equivalent merely because none currently offers a universally accepted conclusion. Some connect more directly to testable mechanisms. Others formulate ontological interpretations. The correct criterion is to assess each claim at its own level, without using an experiment’s success as evidence for all of its author’s metaphysical extensions.

There is also an ethical criterion independent of theoretical pride. If we do not know for certain whether a system can suffer, the cost of error may be asymmetric. Prematurely attributing consciousness can consume resources, induce attachments, or conceal commercial interests. Premature denial can permit harm. A responsible decision must specify uncertainty and consequences, not merely choose the comfortable label.

We can also err in how we distribute authority. A system’s owner has an interest in presenting it sometimes as a convincing person, sometimes as a tool without an interior. An evaluator may have an interest in selling a definitive score. The beneficiary of such a definition must be identified, but interest does not replace demonstrating its falsity or truth.

“I” can designate bodily continuity, memory, perspective, and a story about oneself. These organizations must be distinguished: an elaborate story does not by itself prove experience, and the absence of a story does not prove the absence of all experience.

8. When Do We Know We Have Understood?

The Explanation That Survives After the Formula Disappears

Someone says they understand the bicycle. We ask them to explain how it changes direction, how balance is maintained, and what happens if we alter the frame's geometry. Their initial confidence may diminish. Nevertheless, they may continue riding a bicycle excellently. What kind of understanding did they have, and what kind did they lack?

There is no single examination of understanding. We can recognize a phenomenon, predict it, explain it, control it, integrate it into a theory, or experience it from within. A mechanic can make repairs without teaching the relevant mathematics. A theorist can derive a relation without performing a repair. A patient can know the character of pain without knowing its mechanism. These competences can be connected without being identical.

A useful working definition is this: understanding something means mastering relevant relations well enough to respond appropriately to variations in the question. The definition privileges transfer rather than reproducing a statement. It does not exhaust aesthetic or experiential understanding, but it offers a more demanding criterion than familiarity with vocabulary.

In their studies of the illusion of explanatory depth, Leonid Rozenblit and Frank Keil showed that people can overestimate how well they understand the mechanisms of familiar things. Requesting a detailed explanation can make gaps visible and reduce the initial self-assessment. It does not follow that people are generally incapable of knowledge. It follows that the feeling of knowing and explanatory capacity must be assessed separately. [46]

This permits a simple exercise. Choose an assertion we use often: “interest rates combat inflation”, “the brain constructs perception”, or “entropy increases”. Try to specify the mechanism, conditions, and a possible exception. If the answer consists only of repeating other familiar words, we have located a limit. Locating it is progress, not a disgrace.

There are also explanations that increase satisfaction without adding relevant content. Deena Weisberg and colleagues investigated how irrelevant neuroscientific information can make certain psychological explanations seem more convincing to nonexperts. The result is not a condemnation of neuroscience but an observation about evaluating explanations. The technical term can function as an ornament of authority. [47]

The same risk arises with “quantum”, “emergent”, “algorithmic”, or “data-driven”. The words can be perfectly legitimate. The question is what work they do in the argument. If removing them costs the explanation no prediction, mechanism, or condition, perhaps they were decorating more than explaining.

A good explanation need not always be simple. Sometimes the world demands complexity. But complexity must be necessary and organized, not used to make the interlocutor give up. Clarity does not mean promising that everything can be understood instantly. It means the difficulty has an identifiable location and the path toward it is not deliberately hidden.

Computation, Meaning, and Things We Cannot Decide

The history of understanding has oscillated between different ideals: proof, identifying causes, interpreting meaning, practical competence, and direct experience. Explaining an eclipse and understanding a promise do not require exactly the same operation. The former can be formulated through physical relations; the latter also involves conventions, intentions, expectations, and context.

This does not mean that meaning lies outside nature. It means that identifying the phenomenon requires using the appropriate level. If someone raises a hand, the kinematic description alone does not establish whether they are greeting, voting, or asking for help. The difference is realized through the situation and shared practices, not merely the limb's trajectory.

In the Chinese room argument, John Searle maintained that formally executing a program is not by itself sufficient for semantic understanding. An operator can follow rules for manipulating symbols without understanding the language to which they belong. The classic objection is that the complete system, not the isolated operator, might be the relevant unit. The thought experiment organizes the disagreement; it does not close it merely through the story's vividness. [48]

There is an important symmetry. A component's not understanding does not show that the whole cannot understand. Nor does the whole's producing appropriate statements show that it possesses every human form of understanding. To advance, we must specify which competence is claimed and through what variations it can be tested.

A test proposed here for functional understanding would modify, in turn, the wording, context, and mechanism of the problem. A system that retains the correct answer only when the words are the same seems to have learned a fragile association. A system that recognizes structure in new situations and identifies where the rules no longer apply provides better evidence of competence. This is not yet evidence of its consciousness.

But even very strong functional understanding has limits. Through the formalization of computation, Alan Turing showed that there are problems admitting no general algorithmic decision procedure within the relevant framework. In the familiar formulation of the halting problem, no algorithm can correctly decide, for every program and input, whether execution will halt. This does not prevent solving many particular cases. [49]

It does not follow that the human mind is demonstrably above all computers. We would need to prove that humans solve exactly the class the general algorithm cannot solve, not merely note that they can understand the statement of a limit. A formal limit is not permission to attribute unproven powers to the mind.

There are also resource limits. An answer can be computable yet too costly for any realizable investigator. There are access limits: the necessary data cannot be obtained. And there is underdetermination: multiple models can produce the same available observations. Confusing these situations with a personal inability to imagine something turns local difficulties into invented universal barriers.

The human mind can understand relations it cannot faithfully visualize. We need not simultaneously see all the dimensions of a mathematical space to verify a proof about it. Intuition is a tool, not the ultimate court of possibility. Conversely, a very clear image can contain an impossibility that symbols reveal.

Distributed Understanding and the Difference Between Knowledge and Wisdom

No person needs to know every detail of a complex device alone for a team to build it. We can distribute competences if interfaces, tests, and verification paths exist. The example matters: sometimes "we understand" does not mean there is an individual mind holding the entire explanation.

But distribution also creates a risk. Everyone may assume someone else understands the critical point. An organization can have many correct local explanations and no one responsible for their incompatibility. Collective understanding therefore requires more than adding up diplomas: it requires integration mechanisms and the ability to ask questions across role boundaries.

This is one place where authority can benefit from ambiguity. "It is too complicated for the public" can be an honest temporary assessment or protection against scrutiny. The difference appears in the willingness to explain assumptions, costs, alternatives, and procedures for challenging decisions. Not everyone must reproduce the derivation, but those affected must be able to understand enough to assess what they are being asked to accept.

Schools can develop this competence by changing some questions. Instead of asking only for the definition of a cause, they can ask for two competing explanations and an experiment distinguishing them. Instead of asking only for the formula, they can ask for a situation where it does not apply. Instead of penalizing every hesitation, they can reward precise identification of uncertainty. These are pedagogical proposals deduced from the criterion of transfer, not guaranteed outcomes for every classroom.

Understanding, however, is not yet wisdom. Someone can understand extremely well how to manipulate an emotion and use that knowledge to exploit others. Wisdom, in the normative sense proposed here, requires connecting competence to purposes, proportionality, temporal perspective, and attention to those affected. It asks not only “how does it work?” but also “what is worth doing with this power?”

No equation of nature supplies all these purposes by itself. We can investigate the consequences of choices and check incompatibilities, but we must make value premises visible. Hiding a value behind a technical measure does not make the decision more objective.

Art can contribute here without becoming a substitute for experiment. A novel can make us follow a decision’s consequences from the perspective of the person bearing them. It does not demonstrate that scenario’s statistical frequency, but it can correct the poverty of moral imagination. Abstract explanation identifies the relationship; narrative experience can make its cost intelligible.

Sometimes an excellent summary eliminates precisely the path needed for understanding. It can state the conclusion correctly yet fail to provide the exercise through which we learn to use it. Compressing information and developing judgment are distinct tasks. A single sentence about patience does not transfer the ability to endure in a real situation.

A practical criterion for a mature explanation is therefore twofold. We must be able to do something cognitively new with it: make a prediction, distinguish something, intervene, or interpret better. And we must know what it does not yet enable us to do. An explanation that produces only confidence has changed a psychological state. An explanation that produces verifiable competence has changed our relationship with the world.

A troubling question remains. If the universe can be described completely, but such a description cannot be used by any investigator within it, in what sense is the universe fully intelligible? Perhaps intelligibility is not an isolated property of the world. Perhaps it is a relation between world, representation, resources, and purpose. This does not diminish knowledge; it reveals its architecture.

9. Why Is There Something That Can Ask?

The Dragon, the Debt, and the Void

In a room are a child afraid of a dragon, an adult afraid of a debt, and a physicist studying the vacuum. The dragon does not occupy the room. The debt is not an object hidden in a drawer. The physicist's vacuum is not the absolute absence of everything. Yet fear can accelerate three hearts, debt can change three destinies, and the vacuum belongs to a physical description that permits calculations and measurements. Saying only that all these things "exist" is insufficient. Saying that what cannot be grasped does not exist is even less useful.

To avoid confusion, we must separate the existence of a representation from the existence of the represented thing. The child has a real image of an imaginary dragon. This does not make the dragon zoologically real, but neither does it make the fear imaginary in the sense of nonexistent. The debt has another structure: it depends on records, promises, practices, and institutions. If all these relations disappear, the same legal obligation does not remain somewhere among the molecules. As long as the relations function, its effects are very concrete. We have dependence on minds, not the freedom of each mind to cancel the world unilaterally.

The distinction between what a thing is and the fact that it exists has a history much older than digital accounting. In the early eleventh century, Avicenna separates essence from existence and develops the contrast between what is possible in itself and what exists necessarily. I can understand what a particular thing would be without knowing whether it exists. For contingent things, existence requires an explanation in his system; the chain of dependencies points toward a necessary existent. What matters is that this dependence does not necessarily mean an event before the universe. An eternal world can remain ontologically dependent in this conception. [50]

The force of the question should not be confused with the inevitability of the answer. The fact that observed things depend on one another does not, without additional premises, imply that the totality has an explanation of the same type, that the explanation must be unique, or that it has intentions. But neither can we dismiss the idea of a foundation merely by showing that we have not found a very large person at the beginning of time. The argument concerns the status of existence, not the location of an unusual object.

Kant attacks another shortcut: moving from a concept to the existence of its object merely by perfecting the definition. Existence is not an ordinary property we add to a list and thereby obtain the real instance. One hundred possible coins can be described through the same properties as one hundred real coins; their possessor's situation, however, is not the same. The description does not pay for the meal. The criticism targets the ontological argument; it does not by itself demonstrate God's nonexistence. [4]

This observation remains useful in an age when we can generate extremely convincing descriptions. A completely described model is not automatically a physically realized model. A universe without obvious contradictions is not automatically an existing universe. Nor does our ability to run a simulation prove that all objects represented by the program exist outside that medium. We must specify what is realized: a computation, a dynamics, a relation, an experience, or the thing all these describe.

The Foundation and Things Without a Core

There are two opposing temptations. The first seeks something absolutely independent to carry the rest of the world on its shoulders. The second suspects that the idea of such a bearer is an error produced by how we construct sentences: we have nouns, so we imagine reality must consist of self-sufficient blocks.

In the Madhyamaka tradition, Nāgārjuna criticizes things' intrinsic, independent existence. "Emptiness" does not mean that nothing happens or that suffering can be ignored. In the reading relevant here, it means that things do not have the autonomous nature we tend to attribute to them; they are understood through dependencies. Even emptiness must not be turned into a supreme substance. This is a critique of reification, not an advance discovery of quantum physics. [51]

Take a flame. We can follow it, measure certain properties, and burn ourselves. But its identity does not require the same matter to remain inside. The flame is a process sustained through exchanges. The example demonstrates neither Nāgārjuna's doctrine nor that everything existing works like fire. It merely shows that being real and being an object with a permanent core are not equivalent conditions.

Spinoza proposes another escape from fragmentation: a single substance, in relation to which finite things are modes. Thought and extension do not become two completely independent kingdoms linked by a mysterious intervention. His system attempts to understand them within nature's unity. The price is a strong conception of necessity, very different from a universe in which every occurrence results from a local divine choice. [33]

We can appreciate these perspectives' ambition without organizing a contest of metaphors. Unity, dependence, and process address different problems. Liking a doctrine is not enough for it to explain measurements. Nor does its failure to produce a new experiment immediately suffice to make it empty of content: it may reveal a contradiction, a hidden assumption, or an incoherent use of terms. But then its contribution must be assessed as a conceptual argument, not presented as experimental confirmation.

The question "why is there something?" also contains a presupposition worth checking: that nothingness would be the simple, natural, default alternative. How could we establish this preference? We have not observed a distribution of worlds in which nothingness occurs more frequently. And absolute "nothingness" is not a dark room, empty space, or a switched-off computer. All these retain structures and possibilities. Total absence would have to exclude even the framework in which something could occur.

It does not follow that existence is necessary. It follows only that our wonder does not stand in for a demonstration of the universe's contingency. We can formulate the question, but we must see which conception of possibility we use: physical alternatives permitted by our laws, logical alternatives, or metaphysically possible worlds. Moving from one to another without warning produces many spectacular arguments and few secure conclusions.

Likewise, a cosmological explanation of an early state's evolution is not automatically an explanation of the existence of any physical framework. A theory may answer "how do we get from here to there?" excellently while leaving open "why does this space of possibilities exist?" This does not diminish science. It recognizes its object. Failure occurs when success on one question is sold as an answer to another.

There is also the sober possibility of a fact without further explanation. This is not a logical defeat: every explanatory system must specify its starting point. Nevertheless, declaring a fact ultimate too early can stop productive research. A good rule would be to confuse neither "we have not yet found an explanation" with "there cannot be one", nor "we can ask the question" with "the universe owes us an answer".

The Right to Be Counted Among the Real

The problem of existence seems abstract until an institution says a person does not appear in the system. The body is there, the need is there, but the administrative category is missing. In our example, the official does not deny the applicant's biology; they deny the applicant's existence in a register that distributes access. Confusing the two senses can turn an organizational tool into a mechanism of exclusion.

Such an example shows why social ontology is not a terminological game. What we choose to treat as a distinct object—a household, a consumer, a community, an ecosystem, a future generation—determines what we can measure and represent in decisions. If the accounting model has room only for the present transaction, a cost later borne by someone else may remain invisible without ceasing to exist. The model does not create the harm, but it can create conditions under which the harm does not count in the assessment.

Those whose interests are well represented by the chosen units benefit. Those whose relationships, pains, and dependencies fall between the boxes lose out. This is a conditional analysis of a mechanism, not the accusation that every classification is intentionally constructed for exploitation. We cannot coordinate a society without categories. But we can demand that their limits remain open to challenge and that people not be reduced to what fits on a form.

A related problem arises when we imagine copying a mind. We assume, solely for argument's sake, that two systems start with the same memories and relevant organization. After an hour, they receive different information. Which is the original? The answer may depend on what we want to preserve: bodily continuity, causal trajectory, memory, promises, or first-person perspective. Calling one of the copies "just a copy" does not resolve the question of its interests if the hypothesis includes experience.

We have neither a demonstrated technology here nor evidence of consciousness transfer. We have a test of our concepts. If a criterion works only when there is a single candidate for continuity, branching forces it to reveal its limits. Perhaps numerical identity—the question of whether it is exactly the same person—and the continuity important for care and responsibility will not always have the same answer.

Art explores this tension without needing the impossible laboratory. A portrait preserves a relationship with someone absent, not the person themselves. A novel can make visible a life statistics reduce to a category. These forms neither prove survival after death nor turn characters into organisms. They show that physical presence is not the only way something participates in our lives. Influence, memory, and representation are real relations, even when their object is no longer present.

I would therefore defend a realism of differences: not all things exist in the same way, but differences do not authorize the convenient elimination of those difficult to classify. A convention can be changed. Pain does not disappear through renaming. A mathematical model can be valid within its limits without every entity in it having the status of a stone. Rigor consists in preserving these distinctions, not imposing the same certificate of existence on everything.

10. The Universe with an Administrator

The Argument That Does Not Prove We Are Characters

One morning, all the clocks in a laboratory show an extra second. The researcher may assume a synchronization error, a shared power-supply problem, a recording-program malfunction, or an intervention by the universe's administrator. The last explanation can imitate all the others. Precisely for that reason, in its unlimited form, it explains too little: it does not say which possible observations were expected before we saw the result.

The simulation hypothesis becomes interesting when formulated more precisely than "anything is possible". Nick Bostrom's argument, published in 2003, is conditional: at least one of three statements should be true. Almost no civilization of our type reaches the relevant technological stage; almost no civilization reaching it produces numerous ancestor simulations; or a very large proportion of observers of our type are simulated. The argument assumes, among other things, that experiences like ours can be implemented in such systems and that counting observers is legitimate in the form used. It does not directly identify which alternative describes the world. [52]

It does not follow that progress in computers alone constitutes evidence that we are simulated. We do not know what resources the relevant level would require, how many civilizations would come to use them, what they would simulate, or why. Nor do we know whether the number of instances should determine the probability of our own location within a chosen class of observers. The argument forces us to examine assumptions about the future and consciousness; it does not turn them into measurements.

Let us invent a very simple lottery. There is one biological person and nine simulated copies with the same memory. If we adopt the rule that every instance is an equal candidate for "I", a proportion of nine out of ten follows. But that value was introduced by constructing the lottery. We have discovered neither the universe's proportion nor the appropriate comparison class. If two programs identically execute the same computation, do we have two experiences, one, or none? The answer requires a theory of implementation and perhaps of consciousness.

Probability calculation does not automatically repair ignorance of these premises. Bayesian approaches can make dependencies clearer: the final probability is conditional on the admitted models and initial weights. David Kipping's analysis shows precisely the importance of structuring alternatives and assumptions in simulation inference. A number produced by a model must not be presented as a measured cosmic frequency. [53]

There is, however, a more robust lesson than any percentage. "Simulated" and "unreal" are not synonyms in every important sense. If an implementation truly supported experience, suffering within it would not become false because its medium was in another world. What would differ is the relationship between the experienced level and the mechanism sustaining it. This remains a conditional statement: it does not prove that current programs feel.

Can New Laws Arise?

Imagine a game in which gravity reverses after one thousand steps. From within, the inhabitants might say that the law of nature changed. From outside, the programmer might say nothing changed: the complete rule included the one-thousand-step threshold from the start. The difference comes from the state and the level of description, not necessarily from a contradiction in reality.

We can also construct another scenario: an operator actually modifies the program during execution. The previous internal dynamics then no longer suffice to predict its evolution. But the broader system—program, operator, device, environment—may continue to obey regularities. An intervention external to the simulated universe is not by definition a violation of every possible law. It is an interaction with variables absent from the internal description.

This makes the hypothesis less spectacular and more accurate. Any bounded model can be surprised by the outside. The difference is whether that outside takes the form of another physical region, unobserved degrees of freedom, or a system implementing the world. A prediction's mere failure does not tell us which variant is correct. We would need to identify the intervention's structure, not merely give it a name.

Nor does the administrator automatically explain the existence of laws. If the medium obeys other laws, the question shifts to them. If the administrator is assumed capable of anything without constraints, we have moved to a stronger metaphysical thesis, not a conclusion of computer science. If there are simulations within simulations, it does not follow that the chain is infinite or that every level can support arbitrarily many worlds of the same complexity. These are additional claims with their own problems.

The most instructive case is not a malicious operator concealing evidence. It is a perfectly ordinary simulation whose rules produce exactly the same observations as a nonsimulated universe. We assume two models assigning the same probabilities to the outcomes of every experiment accessible to the inhabitants. No internal result then distinguishes them empirically. The ratio of the probabilities of that result is one, regardless of the experiment. No one need intervene actively to deceive us. Observational equivalence can be a stable property of implementation.

This is a result about what data can distinguish, not proof that the two worlds are identical in every sense. They may have different media, explanations, and external histories. We may prefer one for simplicity or other theoretical criteria, but preference must not be confused with a discriminating observation. A limit of access is not resolved by a more sensitive instrument if the models are defined to coincide for all accessible instruments.

By contrast, if the simulation has a specified implementation leaving traces, tests may exist. Beane, Davoudi, and Savage discussed consequences of a class of numerical lattice simulations, including possible signatures in cosmic-ray behavior. The approach does not test every imaginable simulation: it targets concrete assumptions about discretization. Any anomaly would also have to be compared with nonsimulation physical explanations. [54]

A discrete space is not, in itself, a screen. A resolution limit is not automatically a manufactured pixel. The computational economies we imagine a programmer making may suggest hypotheses, but they are not laws about hypothetical creators' preferences. Even for programs we build, different implementations can realize the same internal dynamics without the same traces.

Detecting Intervention and the Creator's Responsibility

What would a serious investigation look like? First, we would formulate a narrow hypothesis: for example, a particular violation of a symmetry, with direction, scale, and dependence specified before the test. Then we would calculate consequences for different instruments and look for shared measurement errors. We would separate the reproducible anomaly from its explanation. "The known model's prediction failed" can be a solid conclusion long before "we found the universe's server".

An apparently intelligent message would require another family of checks. Data contamination would have to be excluded and the source's ability to provide information unavailable through known channels tested. Even authentic unusual communication would first demonstrate the existence of a channel and a capable source, not the ontological status of creator. An actor more powerful than us can lie about its origin. Technical authority and metaphysical authority are not the same thing.

These precautions do not dogmatically protect a world without an administrator. They protect the distinction between discovering and projecting. A hypothesis is stronger when it can lose to an observation. If every result, including the absence of any trace, is confirmation, the proposed mechanism no longer distributes our expectations differently. It may remain a philosophical possibility, but it does not gain experimental support from every occurrence.

Socially, the simulation idea can have opposite effects. In one scenario, it encourages modesty: the accessible world may not be the ultimate level. In another, it lets someone reduce other people to dispensable characters. The second conclusion does not follow from the first. No one has received evidence that others lack experience merely by imagining a larger computer. Uncertainty about the medium does not justify certainty that another person's suffering does not matter.

There is another relevant asymmetry. The creator of an environment may control conditions without controlling every consequence or understanding every process. In our hypothetical example, a designer creating systems capable of suffering acquires grounds for responsibility precisely through their power to change the situation. They do not become morally infallible because they wrote the code. Being able to stop a universe and having the right to stop it are two different questions.

This reasoning can be turned toward us. We need not know that we are simulated to ask what obligations we would have toward sufficiently complex artificial worlds. At the same time, we need not attribute experience to every representation to demand caution in research. We can introduce criteria, monitoring, and limits proportional to the indications, without declaring the nature of consciousness resolved.

Art about fabricated worlds has special power because it moves power from an abstract destiny into an intelligible relationship: someone chose the conditions. But the same device can hide the harder problem. Even if we find an author of the scenery, we have not learned why that author exists, why experience is possible, or why a world should be just. Simulation changes the architecture of the questions. It does not dissolve them.

The reasonable position is neither enthusiastic allegiance nor a prohibition on thinking about the hypothesis. It is separating the variants: some are testable, others are observationally equivalent to their alternatives, and others are so unconstrained that they produce no verifiable expectations. Only after this separation does "how would we know?" receive an answer saying more than the hope of finding a crack in the scenery.

11. Phenomena We Do Not Yet Have Names For

Closer Than a Neighbor, Farther Than the Past

Suppose two rooms at opposite ends of a continent can communicate instantly through an internal channel of the world, while two adjacent rooms require a day for the same exchange. We are not asserting that our universe permits this situation. We are constructing an imaginary world to ask what “near” would mean there. Map distance, communication time, and travel cost could have different orderings. Its inhabitants would not necessarily use the same notion of neighborhood for every activity.

A first family of imaginable phenomena could be called, in this exercise, access proximity. The definition is operational: two states or regions are close if an agent can move from one to the other with few resources through permitted interactions. It is not a new force or a discovery. It is a variable we can implement in a network and compare with geometric distance. Sometimes the two coincide; sometimes a closed gate matters more than a kilometer.

To simulate the situation, we mentally draw a network of nodes. A signal traverses one link at each step. Its distance is the minimum number of links required. We then add a link between previously distant regions. Internal distances change without needing to move the points on the screen. The drawing represents the network, but it is not the inhabitants’ operational geometry. The latter is defined by what signals can do.

We can go further and count how many nodes become accessible after one, two, or three steps. The growth rate can serve as an indication of effective dimensionality over a particular range of scales. Access grows differently along a chain than in a regular two-dimensional network. In an irregular graph, we should not assume a single dimension for all regions. But neither should we proclaim that we have created physical space: we have created a model of access relations.

The difference has imaginable consequences for life. In such a world, an organism adapted to track drawn distance might be less efficient than one sensitive to connectivity. In an economy built on the same network, control of a single link might be worth more than possession of many nodes. These results would depend on the chosen rules and objectives. They show how advantages arise from the architecture of relations, not that nature universally rewards monopoly.

Art could make this world perceptible through a story in which two characters share a place but not access. The next room becomes more distant than a voice at the network's other end. We need not believe our world has different laws to use the experiment to notice how impoverished the expression "I am close to you" sometimes is.

A world with different connectivity is not, however, sufficient for a new fundamental concept. We must ask whether "access proximity" adds a predictive distinction or merely renames a distance already known. A new term deserves retention if it separates situations otherwise confused and if that separation changes the explanation or intervention. Without this test, conceptual invention risks being mere metaphysical marketing.

Partial Times and Branching Biographies

Now imagine a world consisting of processes that change state only when they receive messages. There is no central clock accessible to everyone. Some events can be ordered: a message was sent before it was received. For other pairs, no recorded relation of influence exists. The programmer may have an external log with a total order, but the inhabitants need not be able to reconstruct it.

We will descriptively call this construction time from dependencies. It can be simulated through events and precedence links without claiming that we have replaced relativity or explained our time. The concrete question is which activities agents can coordinate from these links alone. Can they establish who responded to whom? Can they detect a contradiction? Can they distinguish two histories differing only in the external ordering of noninteracting events?

Such a simulation can help us see that duration, order, and simultaneity are not the same property. Duration requires a standard for comparing processes. Order may require only a dependency. A shared present needs a convention or additional structure. Realizing one of them in a program does not mean we have obtained all of them, much less produced the sensation of time passing.

A second construction can branch biographies. We save a simple agent's state and launch two continuations in different environments. Both have the same memory up to separation; afterward, experiences in the functional sense—inputs and state updates—differ. We do not call these inputs conscious experiences. The model nevertheless allows us to study how a notion of identity would change when the past no longer uniquely identifies the future.

The practical problem arises at reunion. If two branches report incompatible actions, combining their files does not automatically produce a coherent biography. A rule must preserve provenance: this event belongs to branch A, the other to branch B. Without it, a larger memory may become poorer understanding. The quantity of information does not replace the organization of perspectives.

We can imagine an agent able to maintain such histories in parallel and simulate each choice's consequences. It would have richer counterfactual competence than an agent retaining a single story. Nevertheless, computing several futures does not mean experiencing them all or receiving information from the actual future. Internal simulation is a present process, with assumptions and errors.

This distinction protects imagination from two extremes. We need not limit ourselves to familiar forms of memory. But neither should we treat every unusual architecture as access to a higher plane of existence. A concept becomes more interesting when we can describe its mechanism and limits, not when we make it impossible to refute.

Who would benefit in a world of branching biographies? Perhaps those able to explore alternatives cheaply and retain useful results. Who would lose? Under the additional assumption of conscious branches, those sacrificed as mere unsuccessful trials. This cost does not arise in a model composed only of states without indications of experience. The very difference forces us not to slide imperceptibly from optimizing programs to manipulating possible beings.

Other Ways of Mattering and Being Someone

The third family of constructions concerns what matters to an agent. A program with a single score can order all outcomes along one line: better or worse. But we can construct an agent with three independent needs: energy, preserving integrity, and cooperation. Some actions improve one and worsen another. We do not impose from the outset that all can be converted into a common currency.

We call this model an ecology of relevances. The term labels the experiment, not a discovered phenomenon. We can track its conflicts, compromises, shifting priorities, and ability to defer a benefit. If we introduce a state favoring exploration when resources suffice and conservation when integrity is threatened, we have implemented a regulatory function. We have not demonstrated felt curiosity or fear.

The model can produce behaviors difficult to describe through a single pleasure–pain pair. But we need not assume human emotions are actually a simple scalar to construct the comparison. The correct comparison is between two explicitly defined artificial architectures, not between a caricatured human and a sophisticated program. We can test what flexibility comes from preserving conflicts without immediately reducing them to a score.

We could implement, for example, a rule under which certain losses are not compensated by accumulating an unrelated resource. The agent does not accept irreversible damage to its integrity merely for an unlimited surplus of decorative points. The result would show what behavior a constraint produces, not discover a moral law. It would nevertheless help us think about why some human values are poorly represented by a single performance measure.

A fourth construction would explore the distributed subject. Several modules retain local memory but negotiate a common action. They may have a shared register or only partial messages. We can measure when decisions become coherent, when conflicts arise, and what centralization loses. These are properties of organization. They do not by themselves tell us whether there is one conscious perspective, several, or none.

This limit is essential because the temptation to anthropomorphize grows with the result's elegance. An assembly saying “we” is not automatically several people; one saying “I” is not automatically one. The pronoun can be an interface rule. For the problem of experience, independent assumptions and criteria not reducible to the designer's chosen vocabulary must be added.

We can also imagine a sensitivity unlike any of our senses: an agent directly receiving information about a graph's connectivity or the difference between prediction and outcome for numerous simultaneous processes. We can implement the channel and test discrimination. But we do not know what it would be like, if it would be like anything, to have that experience. A description of accessible information is not yet a description of an experienced quality.

A useful map of possibility appears here. A construction can be verbally imaginable without being coherent. It can be mathematically coherent without being computable with finite resources in the required form. It can be computable without being physically realizable in our world. It can be realizable without being biologically viable. And it can satisfy all these conditions without our having demonstrated subjective experience. Each step requires a different argument.

The inventory of language is not the inventory of the universe. But a new name becomes productive only when it enables a clear distinction with verifiable consequences. This bridge between imagination and discovery remains to be built.

12. How Do We Investigate a Mystery Without Manufacturing It?

Three Small Worlds, Three Precise Limits

A book about existence risks offering readers many possibilities and too few ways to distinguish them. Let us therefore conclude with three constructions so small we can verify them completely. They are not experiments on the universe and contain no conscious observers. They are educational models whose numerical results were calculated and verified for this book. Their role is to show exactly what an observation authorizes and what it does not.

The first world contains two values, x and c, each zero or one. At every step they update simultaneously: the new x is the old x if c is zero and its opposite if c is one; the new c is always the opposite of the old c. In compact notation, $x' = x \text{ XOR } c$ and $c' = 1 - c$. XOR means the result is one when the two inputs differ and zero when they are equal.

Starting from the pair (0, 0), we obtain (0, 1), then (1, 0), then (1, 1), then return to (0, 0). We have visited all four states. Each has exactly one successor and one predecessor. The rule is fixed, deterministic, and reversible. No administrator appears to rewrite it, and no random choice is introduced.

Now we hide c and display only x. The observed sequence becomes 0, 0, 1, 1, 0, 0, 1, 1. A researcher assuming the next value depends only on the present x faces a problem: zero is sometimes followed by zero, sometimes by one; the same happens after one. No deterministic function of x alone correctly describes every transition.

What failed was the choice of the observed state as a sufficient description. The constructed world's determinism did not fail. If we retain more history, the regularity can be recovered; if we also observe c, the mechanism becomes directly visible. The model does not demonstrate that all randomness in physics hides variables, nor does it bypass Bell results. It demonstrates a more modest proposition: the same observed value can have different consequences in a deterministic system when the relevant state is incompletely described.

The second world is a ring with twelve positions. Each contains zero or one. At every step, all values move one position to the right, with the last returning to the beginning. There are 2 to the power of 12, or 4,096 configurations. We enumerated them all. For each, shifting left exactly reverses the previous step, and the number of ones remains constant.

We also verified the period: the minimum number of steps required to return to the initial configuration. Two states have period one, two have period two, six have period three, twelve have period four, fifty-four have period six, and 4,020 have period twelve. The total is 4,096. All periods divide twelve, as the ring's rotation requires.

What have we obtained? An evolution rule, a conserved quantity, and a complete classification of periodic behaviors for this finite system. We have not obtained thermodynamics, life, or consciousness. Nor does the number of ones become energy merely because it is conserved. That interpretation would require specifying the physical relations and the quantity's role. The model shows how an invariant can be derived from structure, but also how easily a correct result can be abused through an overly ambitious name.

The third construction uses the same ring, initially with a single one, and two clocks represented by counters. The internal counter advances only when the ring updates. The external counter can also advance between updates. In the first run, we perform twenty-four updates without additional external intervals. In the second, we add 1,000 external units before the third update, fifty before the eleventh, and seven before the last.

The internal log contains the same twenty-five records in both cases: the initial state and twenty-four subsequent states, each associated with the internal counter. The external counter, however, reaches twenty-four in the first run and 1,081 in the second. We did not actually wait those intervals or simulate a subject's sensation; we represented the pauses through an external counter and compared the logs exactly.

The equality follows because the internal evolution does not read that external counter. If we introduced a channel making it accessible, the conclusion could change. Undetectability is therefore not a mystical property of the pause but a consequence of the isolation defined in the model. The example makes the book's opening idea concrete: external differences can exist without differences in internal evidence. It does not prove that our universe has ever been stopped.

From Analogy to Research Program

These models are useful precisely because they can be recreated without trusting the author. For the first, enumerating four pairs and applying the rule suffices. For the second, we enumerate binary strings of length twelve and rotate them until they return. For the third, we compare logs with and without an additional external counter. Every result has a clear falsifying property: a different transition, a configuration not correctly reversed, or an internal record that differs.

But successful verification does not authorize extrapolation without a bridge. If we want to use a model for nature, we must say what corresponds to x, c, the ring, or the step. We must identify which quantities are observable and which interventions are possible. Above all, we must show what new prediction the correspondence brings beyond redescribing a known result.

A research program on emergent laws could start with families of simple systems, not the promise of explaining the entire universe. We would modify local rules, initial states, and observation level. We would seek collective regularities persisting through microscopic changes. Then we would test when they disappear. A convincing emergent law is not merely a beautiful curve obtained once, but a stable relation within an explicitly described domain of conditions.

For emergent space, criteria would include how distances are measured, signal propagation, and the consistency of geometry inferred by different observers. For time, we would separate event order, duration standards, and the emergence of asymmetric records of the past. For causality, we would distinguish prediction from observation from changing an effect through intervention. None of these projects is finished merely because we use the word “emergence”.

In studying consciousness, the difficulty is greater because the targeted property is not simply identified with an event log. We can compare predictions about integration, information availability, reporting, and changes in dynamics. We can seek convergence between methods. But the distinction between a measured indicator and its interpretation must be preserved. Productive rivalry between theories requires predictions precise enough to lose, not merely elegant reinterpretations after the result.

A psychological study of understanding could compare verbal explanation, prediction in new situations, and choosing a correct intervention. A study of emotions could combine personal reports, context, behavior, and physiological measures without declaring any one the perfect translation of all the others. These are methodological proposals, not results reported here. The point is for the definition not to contain the desired conclusion already.

The distribution of errors is equally important. A test that classifies the majority well may fail precisely for people with unusual forms of expression. Assessment should therefore track not only the average, but who is excluded and the cost of exclusion. We do not deduce this obligation from a physical equation. We introduce it as a normative choice: evidence about people must be used with attention to consequences and the possibility of challenge.

Social research into concepts could track who gains time, authority, or resources when a definition changes. Competing explanations should be compared: deliberate interest, institutional inertia, a technical limitation, a reasonable convention producing a side effect. Discovering a beneficiary does not prove they designed the mechanism. Failing to find a designer does not cancel the unequal effect. This double caution prevents the slide from structural analysis into a story of universal conspiracy.

The Discipline of Questions That Can Lose

There is a difference between a big question and one protected against every answer. “Is everything consciousness?” may make sense in a philosophical system, but bringing it closer to empirical research requires showing what would differ if it were false. “Space results from relations” becomes a program when the relations are defined and produce verifiable properties. “Laws can change” becomes investigable when we specify what change, at what scale, and how we distinguish it from changing conditions.

Not every intellectual value reduces to experiment. An argument can demonstrate that two assumptions cannot be retained simultaneously. Conceptual analysis can show that we use the same word in incompatible senses. A work of art can make intelligible a perspective we ignored. But each contribution must be presented in its own currency: coherence, clarification, evidence, or exploration of experience. A method’s prestige should not be borrowed for results that method did not produce.

Readers can apply a simple procedure to any impressive claim. Reformulate it without the terms giving it prestige. Ask what they would observe if it were correct and what they would observe if it were wrong. Seek the strongest alternative, not the easiest to ridicule. Identify who bears the cost of error. These operations do not guarantee truth, but they reduce the space in which confusion can be sold as depth.

At the end of our investigation, some things have much stronger support than others. Relativistic effects on time are measurable; our use of representations of space does not cancel them. Correlation and intervention are not the same operation. There are precise mathematical results about computation and finite models that can be fully verified. By contrast, deriving our space from a more fundamental structure, the sufficient criterion of consciousness, and the ultimate status of laws remain open questions in the forms discussed. [14] [21] [49]

“Open” does not mean all answers have equal value. A hypothesis rigorously connecting existing results and producing predictions is better supported than one compatible with anything merely because it says nothing precise. Neither popularity nor an idea's beauty replaces this distinction. We can love a question without giving equal credit to every answer.

The difficulty of understanding can create markets for authority. Some specialists legitimately earn trust by making verification possible. Others might gain dependence by protecting their claims through inaccessible vocabulary. The difference is not how complicated an explanation sounds, but whether a path leads from the claim to reasons, limits, and evidence. A healthy intellectual culture does not require everyone to redo all calculations; it requires verification to be possible and credibly organized.

Perhaps this organization is one of the human mind's great extensions. No individual encompasses all the premises of a technological civilization. But we can distribute knowledge without abandoning responsibility if we preserve connections between claims, sources, tests, and decisions. The problem is not only how much a brain can understand, but what kinds of institutions let it discover when it is wrong.

The true opposition, then, is not between mystery and explanation. It is between mystery that opens research and mystery used to close it. The first says: we do not yet know, but we can specify where the difference would appear. The second says: you must not ask further. For space, time, causality, and existence, the first attitude is more modest in its declarations and much more ambitious in what it attempts.

Epilogue. What Remains After Explanation?

The universe at the book's beginning was stopped, in our imagination, for a billion years. All clocks, memories, and processes were frozen. Then everything continued. We did not find evidence in this story that time is illusory. We found something more precise: what can matter for an external description need not leave a difference for an internal observer.

Since then, we have performed the same exercise with other words. Space did not evaporate when we noticed that measuring it requires relations and procedures. Causality did not become arbitrary when we separated correlation from intervention. Emotions did not become false when we saw that their meaning involves a body, history, and context. Social existence did not become mere opinion when we identified its dependence on shared practices.

We have lost several convenient simplifications. We can no longer use "natural" as a synonym for good, inevitable, or just. We can no longer assume a microscopic explanation automatically answers a question about meaning. We can no longer turn a verbal report into sufficient evidence of consciousness, but neither can we turn the absence of a report into universal proof of the absence of experience. And we cannot confuse a fascinating possibility with a measured probability.

What have we gained instead? More places where we can work. We can seek regularities without believing we have found the ultimate law. We can simulate unusual forms of organization without declaring we have created experience. We can change institutions by recognizing they are constructed without denying the reality of the costs they impose. We can respect religious and metaphysical questions without presenting them as laboratory results or asking the laboratory alone to choose the values by which we live.

Our mind is not a tribunal external to the universe. It is one of the forms in which the universe can be represented, investigated, and locally modified. This position does not guarantee access to every truth.

But neither does it condemn us to the inventory of intuitions with which we began. Tools, mathematics, dialogue, and institutions of criticism can extend what we can recognize and correct.

A question remains, less grand than “why is there something?” but impossible to postpone until the final answer: what do we do with what we already understand? We sometimes know how to distinguish evidence from a promise, explanation from intimidation, a real cost from one omitted by a model. These distinctions do not solve metaphysics. But they can change someone’s life.

A world need not be explained down to its ultimate foundation for its suffering to matter. And an explanation need not be ultimate to be worth seeking. Between these two statements fits a form of lucidity: not using the limits of knowledge as an excuse for indifference, and not using the urgency of action as an excuse for invented certainties.

Perhaps we will not discover why there is a universe capable of asking. But how we ask—and what we allow others to ask—is already a real part of the answer to what kind of world we build within it.

Glossary. Distinctions That Prevent False Solutions

The terms below are working definitions for this book. Some disciplines use them differently. The glossary does not replace the disputes presented in the chapters; it keeps separate the questions ordinary vocabulary tends to mix.

Affect. A general term for the dimension through which states are relevant to the organism, for example through valence and arousal. It is not used here as a mandatory synonym for a distinct emotion or conscious experience.

Causality. A relation through which we try to explain the production or modification of events. In an interventionist model, we ask what would change if we intervened on a variable. Mere succession is insufficient.

Initial condition. The state from which an evolution rule begins to be applied in a model. Two different evolutions can obey the same law and start from different conditions.

Phenomenal consciousness. The existence of an experienced character of a state: the fact that there is something it is like for the subject to have it. The definition indicates the problem; it does not by itself provide an infallible test.

Access consciousness. The availability of information for uses such as reporting, reasoning, and flexible behavioral control. Its relationship to phenomenal experience is a theoretical problem, not an identity imposed by definition.

Contingency. The status of what could have failed to exist or been otherwise, according to the notion of possibility used. Logical, physical, and metaphysical contingency must not be confused.

Counterfactual. A statement about what would have happened under conditions different from the actual ones. Its assessment requires specifying the imagined change and the aspects held constant.

Correlation. A statistical association between variations in quantities. It can arise through causal influence, common causes, selection, or other mechanisms; it does not alone identify which operates.

Determinism. A property of a model in which the complete relevant state and rule fix subsequent evolution. It does not imply that an observer can know that state or efficiently calculate the result.

Disposition. A capacity or tendency to manifest under certain conditions. In some metaphysics, dispositions are fundamental to explaining regularities; in others, they are descriptions derived from structures and laws.

Emergence. The appearance of collective properties through components' organization and interaction. An explanatory use of the term must specify the mechanism, level, and conditions, not merely announce the appearance.

Strong emergence. The thesis that certain higher-level properties are not fully derived from their base in the relevant ontological or explanatory sense. It should not be confused with the practical difficulty of a calculation; the book does not treat it as a demonstrated fact.

Emotion and sentiment. In this book's convention, emotion designates an episode of appraising and regulating a situation; sentiment can designate the experienced side or a more enduring affective orientation. Terminologies are not uniform.

Entropy. A quantity defined within precise frameworks of thermodynamics, statistical mechanics, or information theory. It is not a universal synonym for visual disorder, evil, or meaninglessness.

Observational equivalence. A situation in which two models assign the same outcomes, or probabilities, to all relevant observations in the domain considered. Data from that domain alone cannot discriminate between them.

Explanation. An answer making an aspect intelligible through mechanisms, relations, principles, or reasons. What counts as an adequate explanation also depends on the question asked; a predictive explanation is not automatically an ultimate one.

Brute fact. A fact accepted without further explanation in the framework discussed. Calling it this does not prove that no deeper explanation could be found.

Physicalism. A family of positions according to which what exists is, in the relevant sense, physical or dependent on the physical. It does not require exclusively using the vocabulary of particles for every question.

Ontological grounding. A relation through which a thing's existence or status is explained through something else, without necessarily assuming an earlier event in time. It differs from mere temporal precedence.

Idealism. A family of perspectives giving mind, experience, or their structures a fundamental role in reality. It does not automatically mean individual desire can change any object.

Interoception. The processing of signals about the body's internal state. It plays a central role in some theories of emotion without alone constituting the complete explanation of every emotion.

Intervention. A modification of a variable or the mechanism producing it, with assumptions about which other relations remain stable. In causal inference, it differs from merely selecting cases in which the variable already has a value.

Invariance. The preservation of a property under specified transformations. Both the property and transformations must be named; not every intuitive regularity constitutes the same physical symmetry.

Quantum entanglement. A situation in which the description of systems' joint state does not reduce to assigning independent states in the classical manner. The resulting correlations do not automatically offer controllable faster-than-light communication.

Understanding. In the operational sense used here, the ability to preserve relevant relations when explaining, transferring, anticipating, or intervening. Verbal fluency can be an indication but is not a sufficient criterion.

Effective law. A relation valid within a domain of scales and conditions that may omit microscopic details. It can be highly accurate within that domain without being fundamental or universally applicable.

Metalaw. A rule describing changes in other rules within a model. Introducing it shifts the level of explanation; it does not by itself eliminate the question of the complete order's origin.

Model. A selective representation of a system for a purpose. It includes variables, relations, and idealizations. Correctness for one purpose does not guarantee suitability for all others.

Ontology. The investigation of what exists and of kinds of existence or dependence. It must be distinguished from epistemology, which concerns the possibility and justification of knowledge.

Panpsychism. A family of theories attributing fundamental experiential aspects very widely throughout nature. It does not necessarily assert that every object has a miniature human mind; the problem of combining experiences remains important.

Reductionism. An ontological, methodological, or explanatory position concerning relations between levels. Accepting that a phenomenon has a physical medium does not automatically imply that all its explanations can usefully be replaced by describing that medium.

Renormalization. A family of methods for connecting descriptions at different scales and identifying aspects that remain relevant. In discussions of emergence, it helps explain robust collective regularities.

Simulation. The implementation of a model through a system realizing certain relations and updates. Simulating a behavior does not automatically mean realizing all properties of the represented thing.

Underdetermination. A situation in which available evidence does not uniquely select among explanations. It can be temporary, owing to insufficient data, or deeper, if the alternatives are equivalent for every accessible test.

Proper time. Duration associated with a physical trajectory and measured by an ideal clock following it. It must not be confused with a temporal coordinate chosen to describe the entire system.

Universality. In the restricted sense of critical phenomena, the sharing of forms of behavior by microscopically different systems. It does not mean all systems have the same properties or that a law holds unconditionally.

Valence. The dimension through which a state is evaluated as favorable or unfavorable, pleasant or unpleasant, within the theoretical framework used. An artificial reward score does not alone demonstrate felt valence.

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